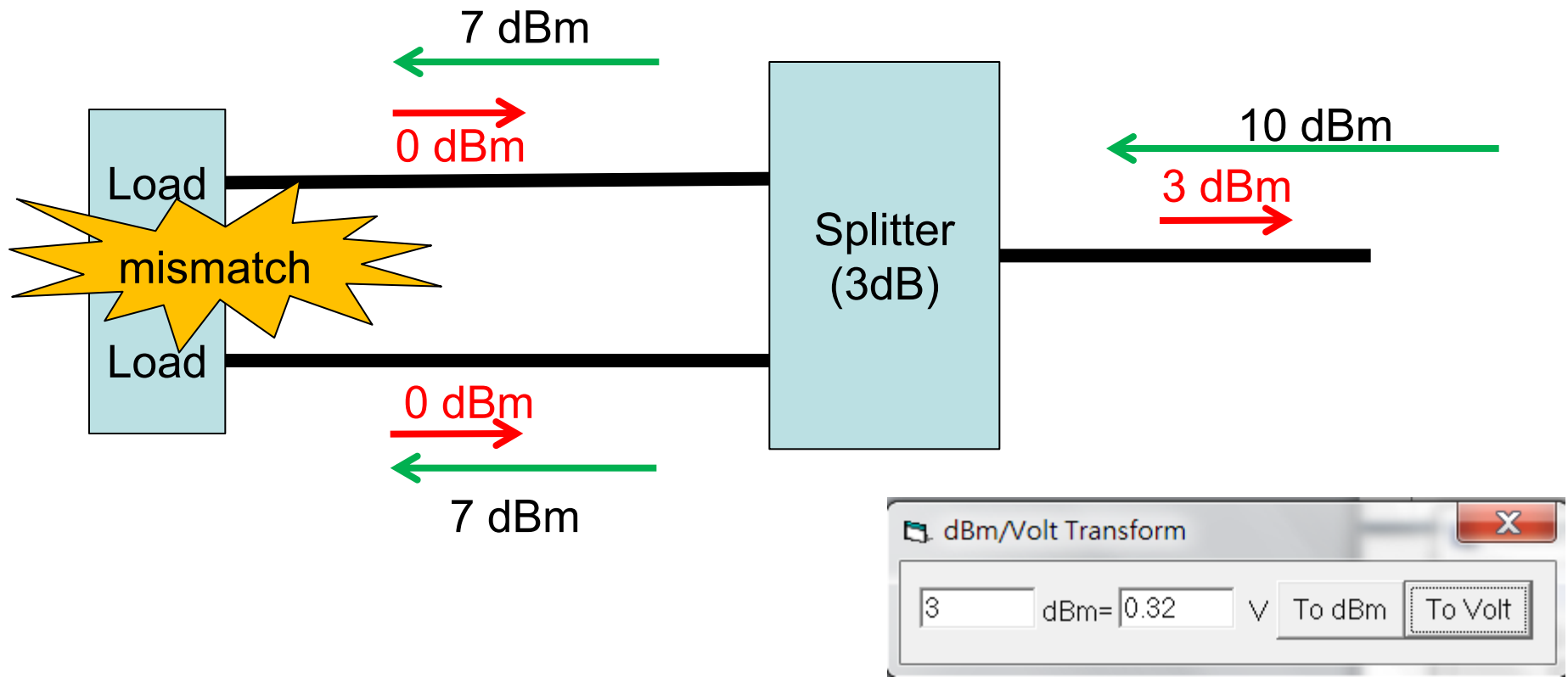


ASUS ZC551KL

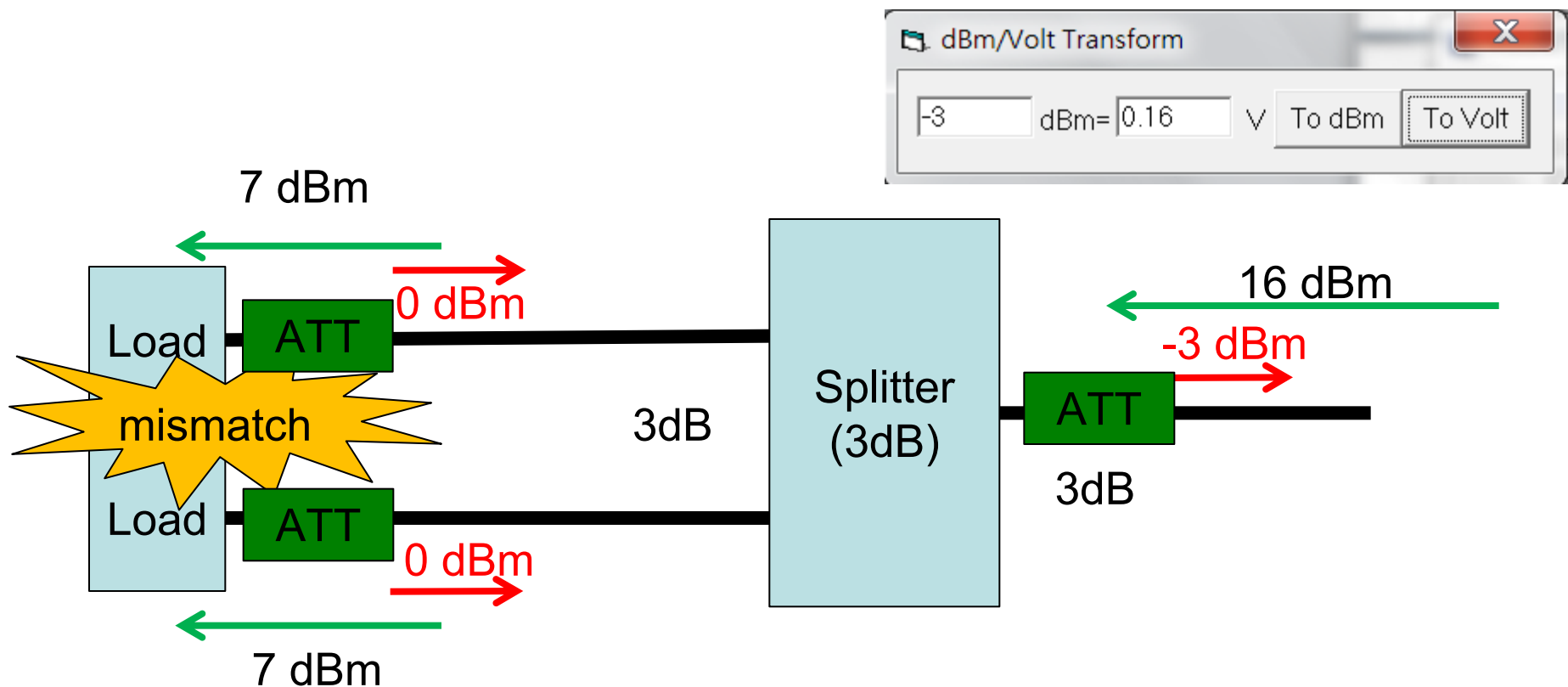
RF Hardware Trouble Shooting Guide

ASUS RF

How to Solve RF System Mismatch Issue (1/2)



How to Solve RF System Mismatch Issue (2/2)



用Combiner接到儀器的一個單port，如圖要補的loss起碼9dB起跳。

Splitter Diff. output port改善6dB isolation。

目前測試run皆以2 port測試的接法，但仍須加3dB衰減器。

ZC551KL RF Areas

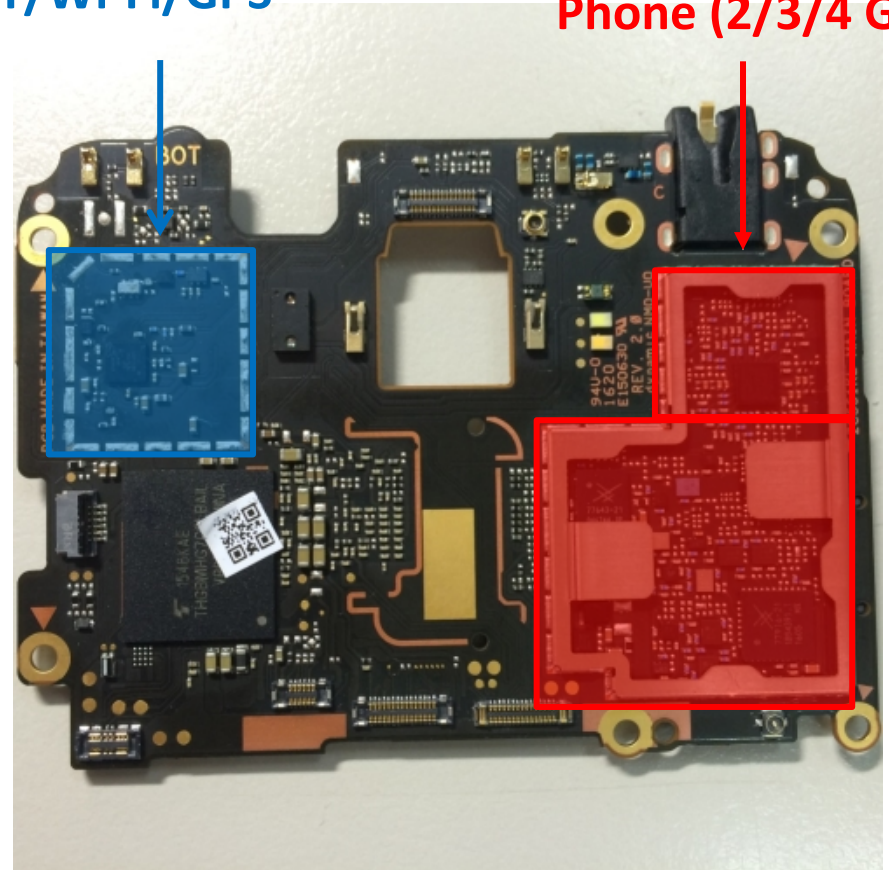
Top side ↓



Bottom side ↓

BT/Wi-Fi/GPS

Phone (2/3/4 G)



ZC551KL RF Supported Bands

TW:

2G: GSM 850/900/1800/1900MHz

3G: WCDMA 850(B5)/900(B8)/1900(B2)/2100(B1)MHz

4G: FDD-LTE 700(B28)/850(B5/6/18/19/26)/900(B8)/
1800(B3)/1900(B2)/2100(B1)/2600(B7)MHz
TDD-LTE 2500(B41)/2600(B38)

WW:

2G: GSM 850/900/1800/1900MHz

3G: WCDMA 850(B5)/900(B8)/1900(B2)/2100(B1)MHz

4G: FDD-LTE 800(B20)/850(B5)/900(B8)/1800(B3)/
1900(B2)/2100(B1)/2600(B7)MHz

IN/ID:

2G: GSM 850/900/1800/1900MHz

3G: WCDMA 850(B5)/900(B8)/2100(B1)MHz

4G: FDD-LTE 850(B5)/900(B8)/1800(B3)MHz/2100(B1)MHz
TDD-LTE 2300(B40)MHz

			PCBA	WW	ID/IN
			Scorpio	Y	Y
2G	GSM		850MHz	Y	Y
			900MHz	Y	Y
			1800MHz	Y	Y
3G	WCDMA		1900MHz	Y	Y
		1	2100MHz	Y	Y
		2	1900MHz	Y	
		4	1700MHz		
		5	850MHz	Y	Y
		6	850MHz		
		8	900MHz	Y	Y
	TD-SCDMA	19	850MHz		
		34	2100MHz		
		39	1900MHz		
LTE	FDD	1	2100MHz	Y	
		2	1900MHz	Y	
		3	1800MHz	Y	Y
		4	1700MHz		
		5	850MHz	Y	Y
		6	850MHz		
		7	2600MHz	Y	
		8	900MHz	Y	Y
		9	1800MHz		
		12	700MHz		
		17	700MHz		
		18	800MHz		
		19	800MHz		
		20	800MHz	Y	
		26			
		28	700MHz		
		29	700MHz		
	TD	30	2300MHz		
		66			
		38	2600MHz		
		39	1900MHz		
		40	2300MHz		Y
		41	2500MHz		

How to distinguish different RF SKUs of ZE551KL (1)

BR/US:

2G: GSM 850/900/1800/1900MHz

3G: WCDMA 850(B5)/900(B8)/1700(B4)/1900(B2)/2100(B1)MHz

4G: FDD-LTE

700(B17/28)/850(B5)/900(B8)/1700(B4)/1800(B3)/1900(B2)/2100(B1)/2600(B7)MHz

方法1: 在有adb.exe檔的資料夾路徑下，輸入cmd按enter叫出command window後輸入以下指令：

adb root

adb root

adb shell cat proc/aprf

得到結果與RF SKU的對照

4=WW 5=IN/ID

6=TW 7=BR/US

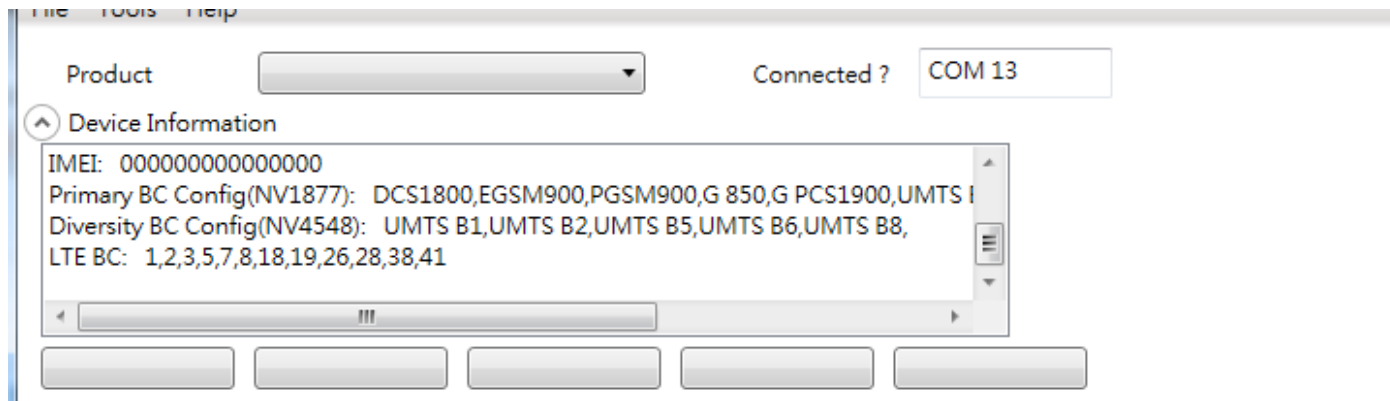
```
C:\Users\prauge\Desktop\Shawn BAT\NUM>adb root
adb is already running as root

C:\Users\prauge\Desktop\Shawn BAT\NUM>adb shell cat proc/aprf
7
```

RF	RF_SKU		
	RF_SKU2	RF_SKU1	RF_SKU0
	GPIO_97	GPIO_87	GPIO_86
	R4203	R4202	R4201
EVB	0	0	0
WW	1	0	0
IN/ID	1	0	1
TW/JP/HK	1	1	0
US/BR	1	1	1
CN(預留)	0	1	1

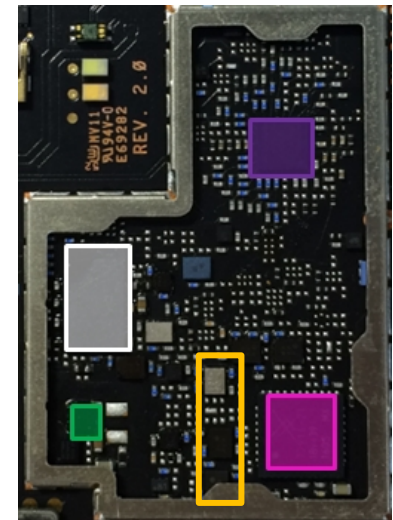
How to distinguish different RF SKUs of ZE551KL (2)

方法2: 在有comport情況下，打開QRCT下的QFDT tool，對照前面提供的supported band判斷RF sku國別

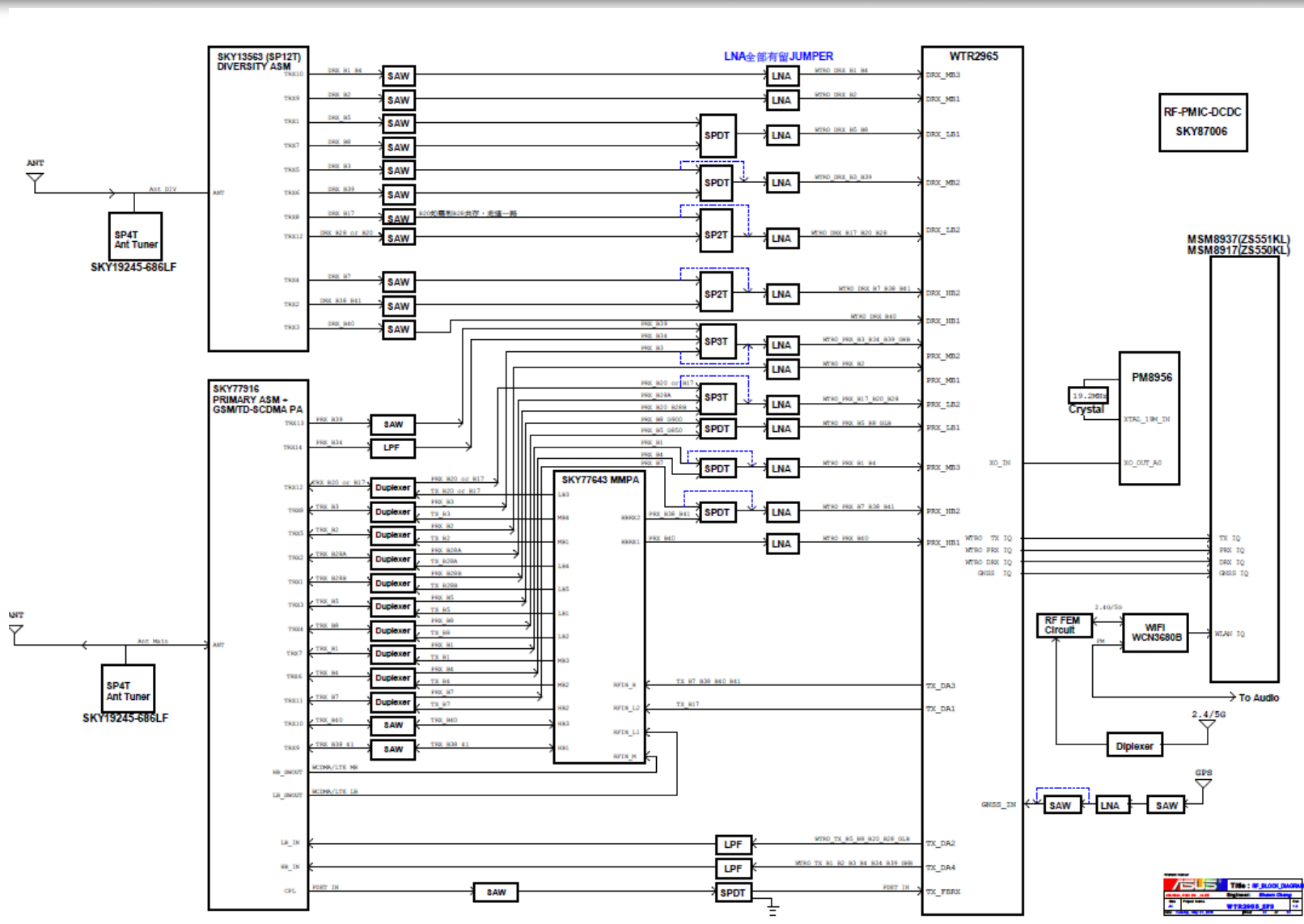


方法3: 打開shielding，如右圖黃框可以看到一行四個duplexer。

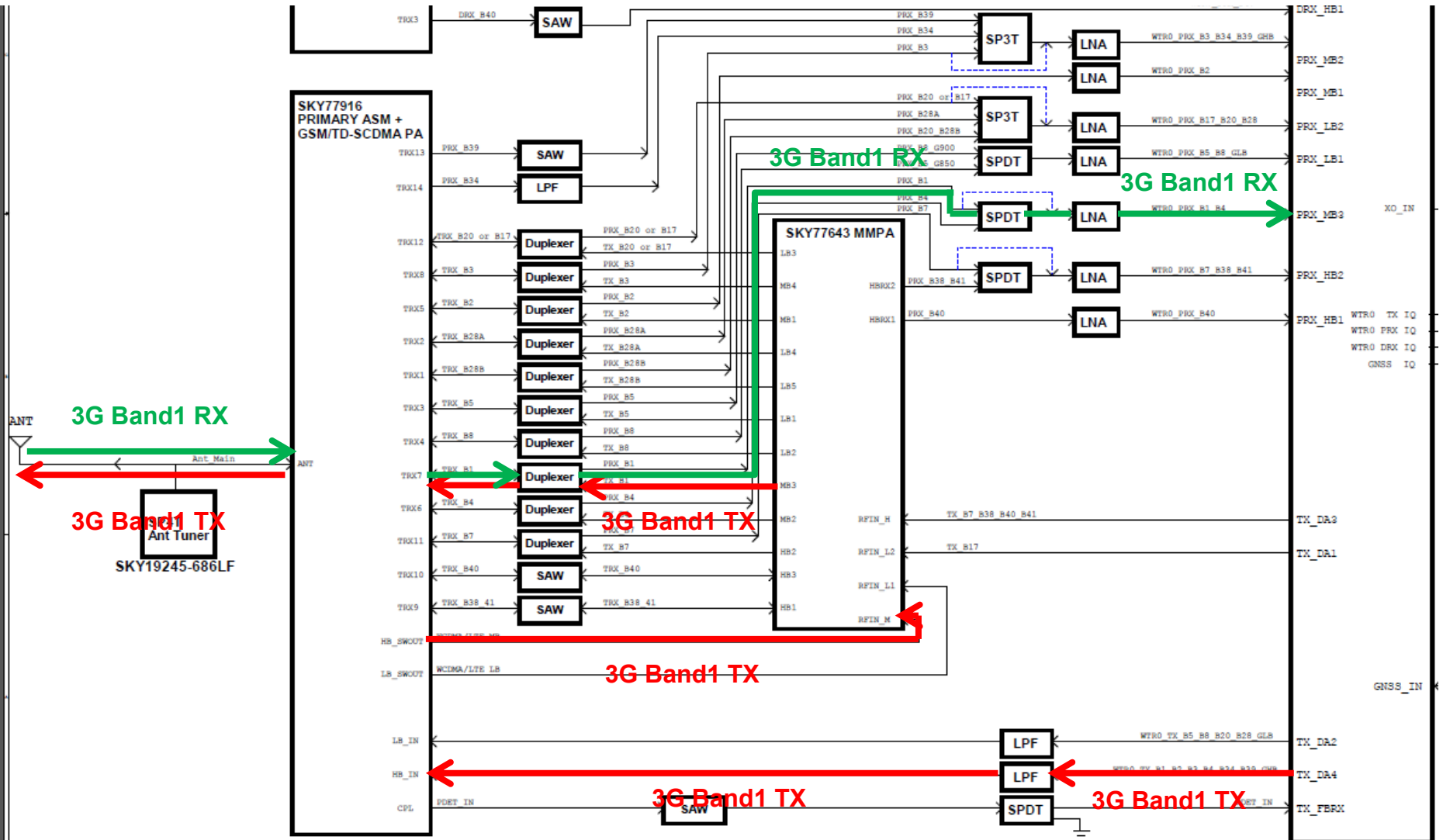
1個是IN/ID，2個是WW，3個是TW/JP，4個是US/BR



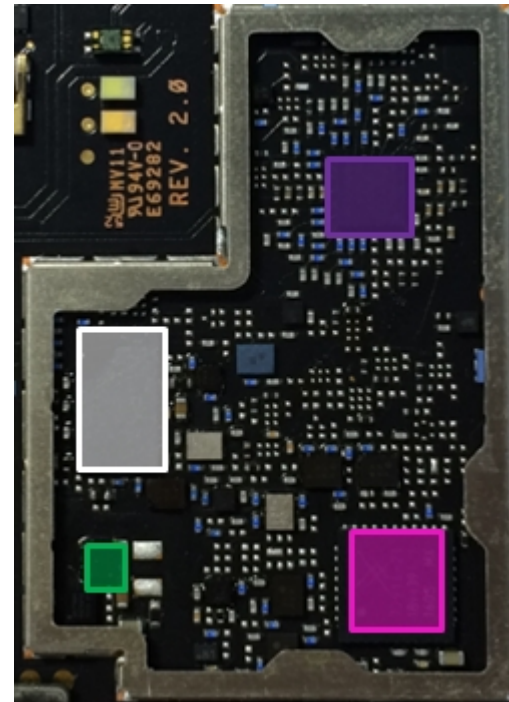
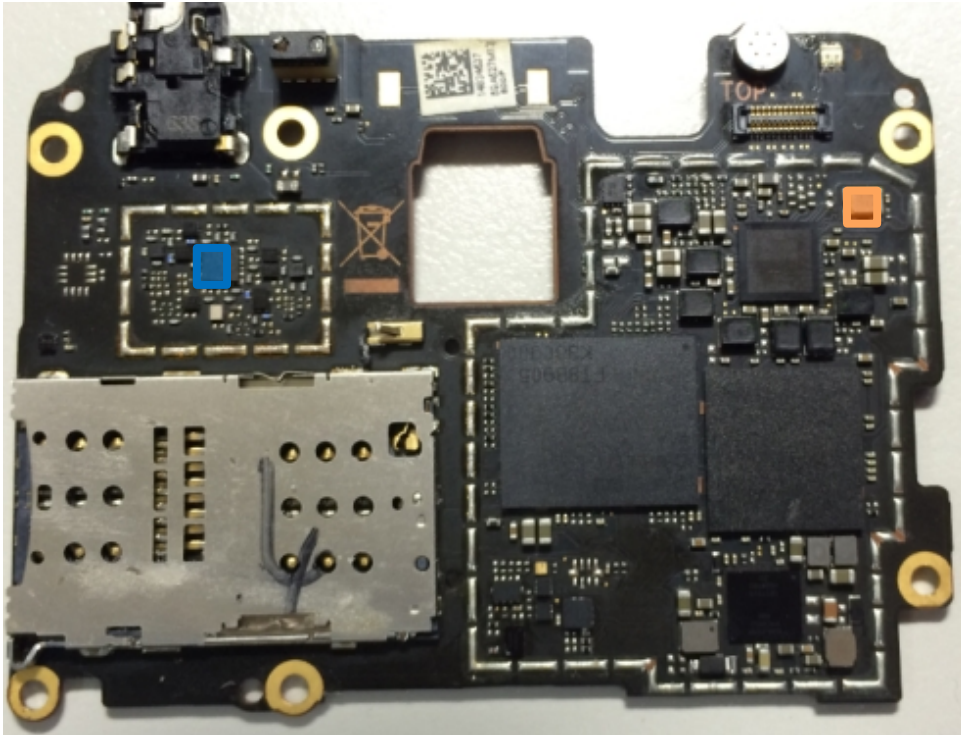
RF Block Diagram : 2G/3G/4G System



RF Block Diagram : 2G/3G/4G System (Ex : 3G Band1)



Key Components : Phone (1/2)



19.2MHz Crystal **X9001**



Main Antenna SW **U4701**



DCDC **U4601**



Transceiver **U4501**

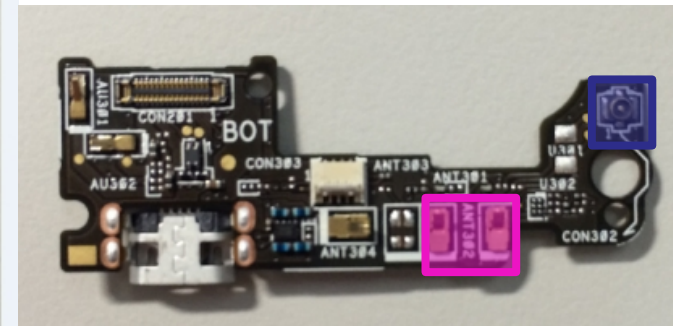
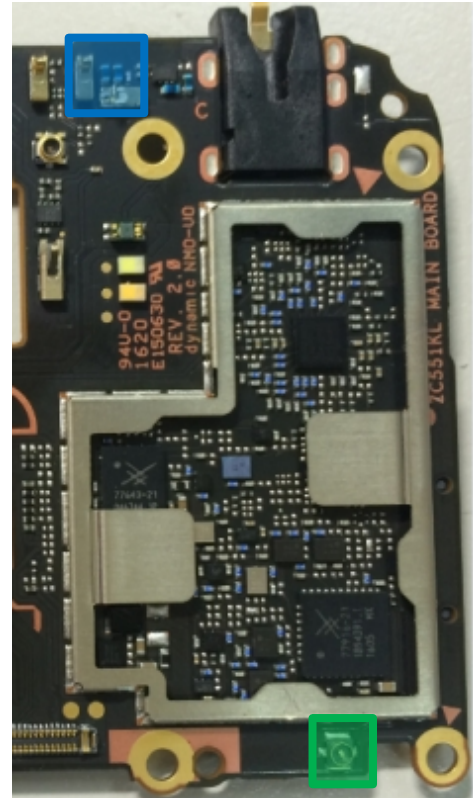
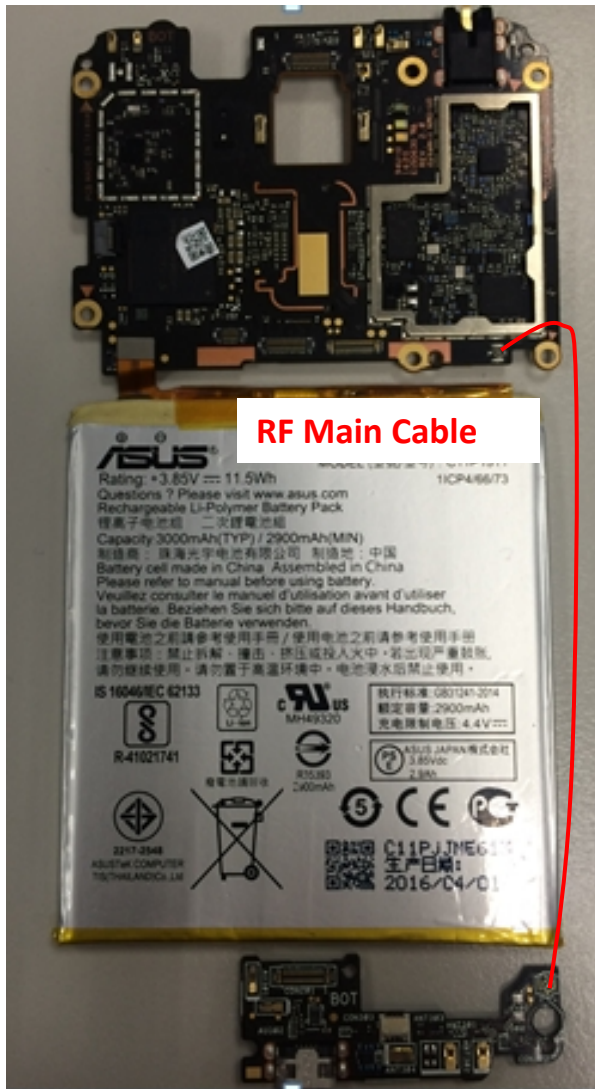


Diversity Antenna SW **U4705**



MMIC **U4805**

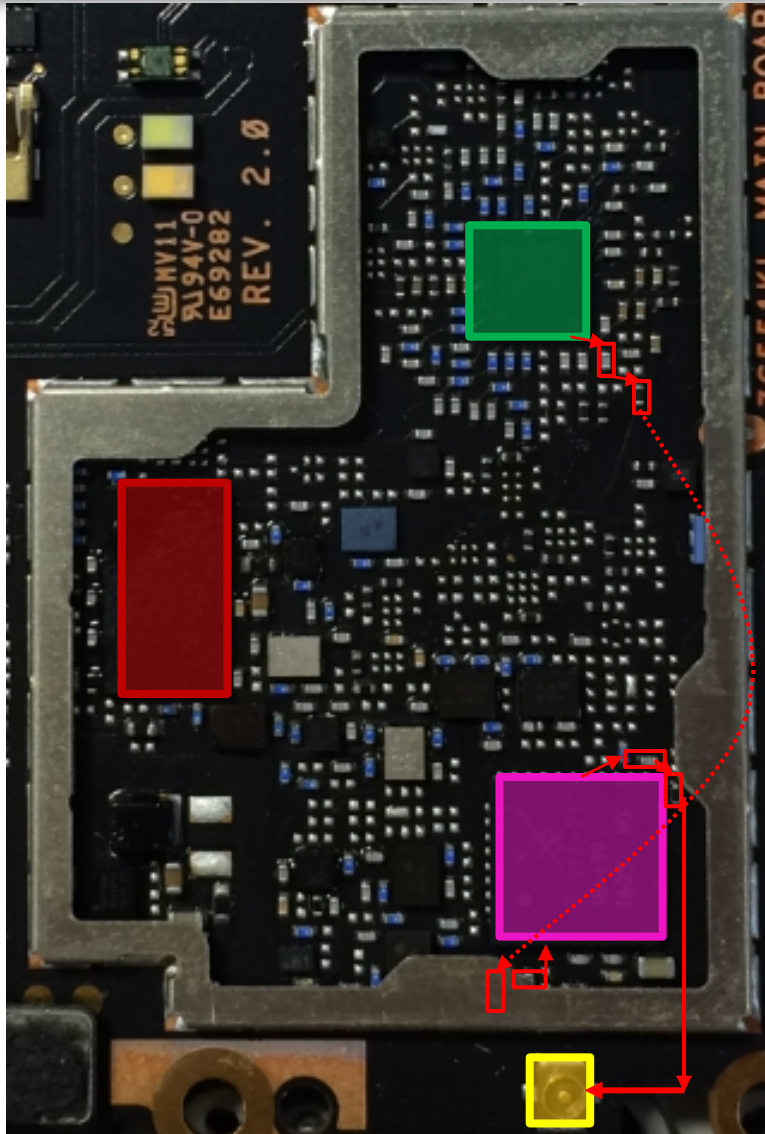
Key Components : Phone (2/2)



- Spring (Phone Div. antenna)
- RF Main Connector
- RF Diversity Connector

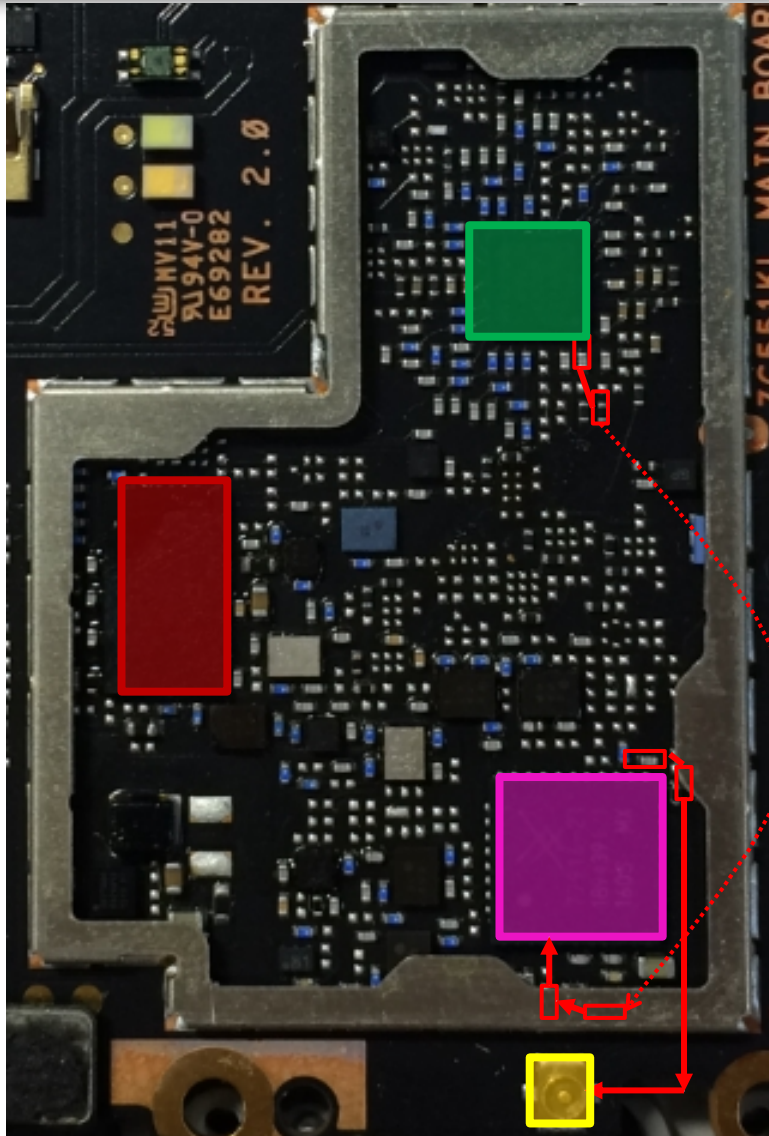
- RF Main Connector
- Spring (Phone Main ant.)

TX Path : 2G GSM850/GSM900



-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**

TX Path : 2G DCS/PCS



Main Antenna SW **U4701**



MMPA **U4805**

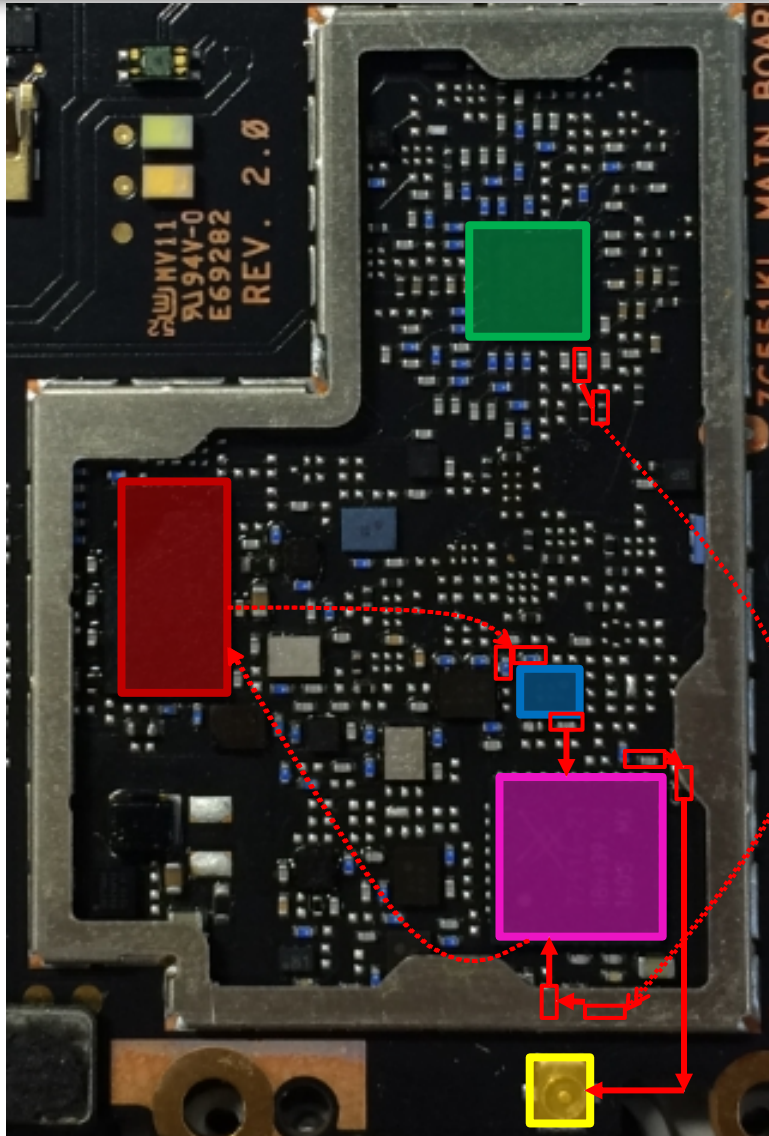


RF Main Connector **CON4701**



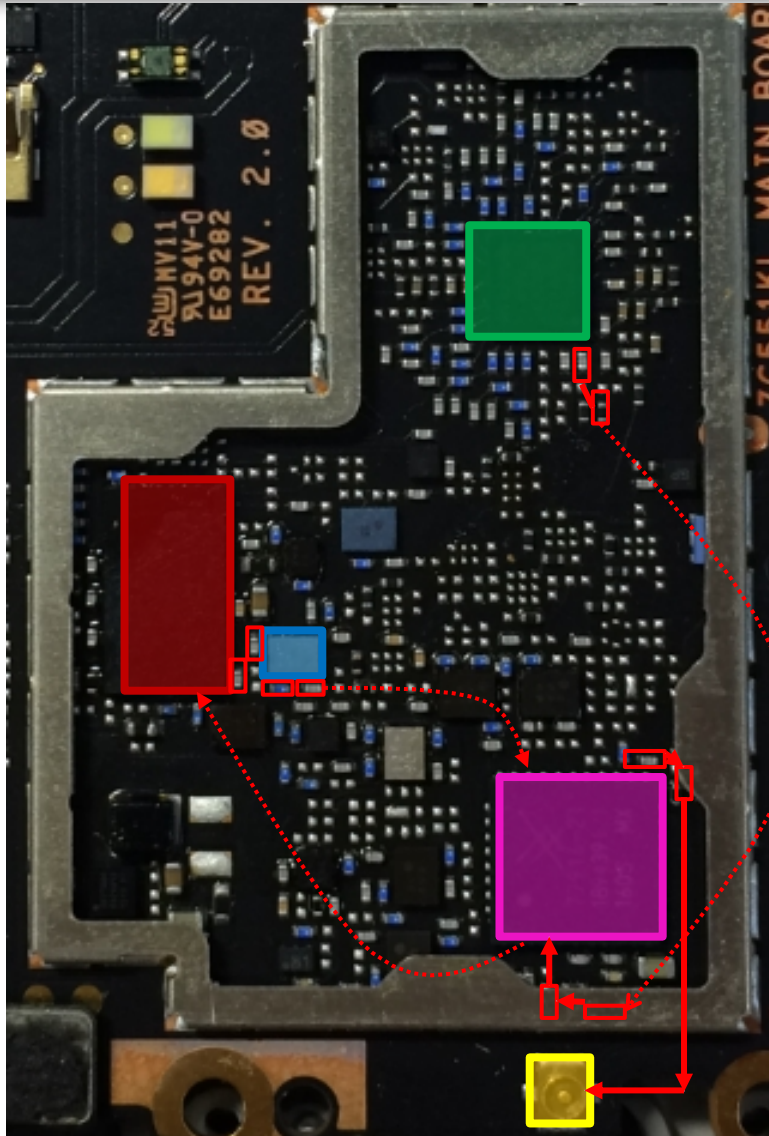
Transceiver **U4501**

TX Path : 3G&4G Band 1



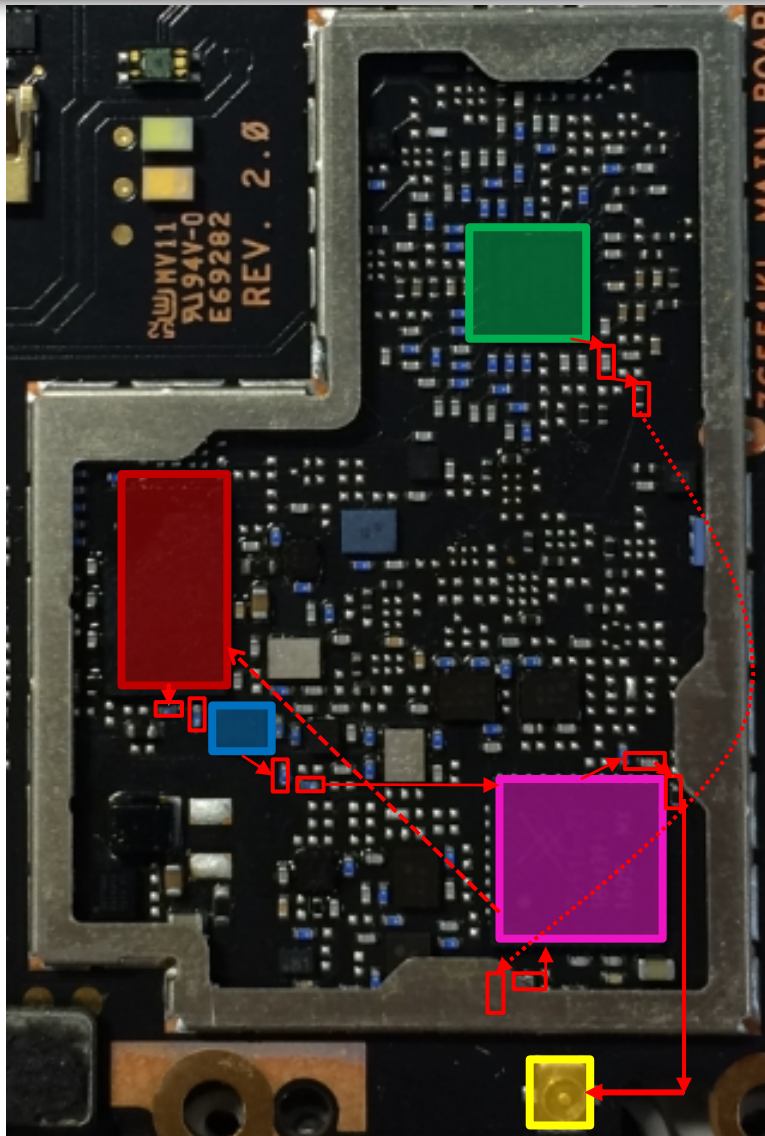
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B1 Duplexer **U4807**

TX Path : 3G&4G Band 2



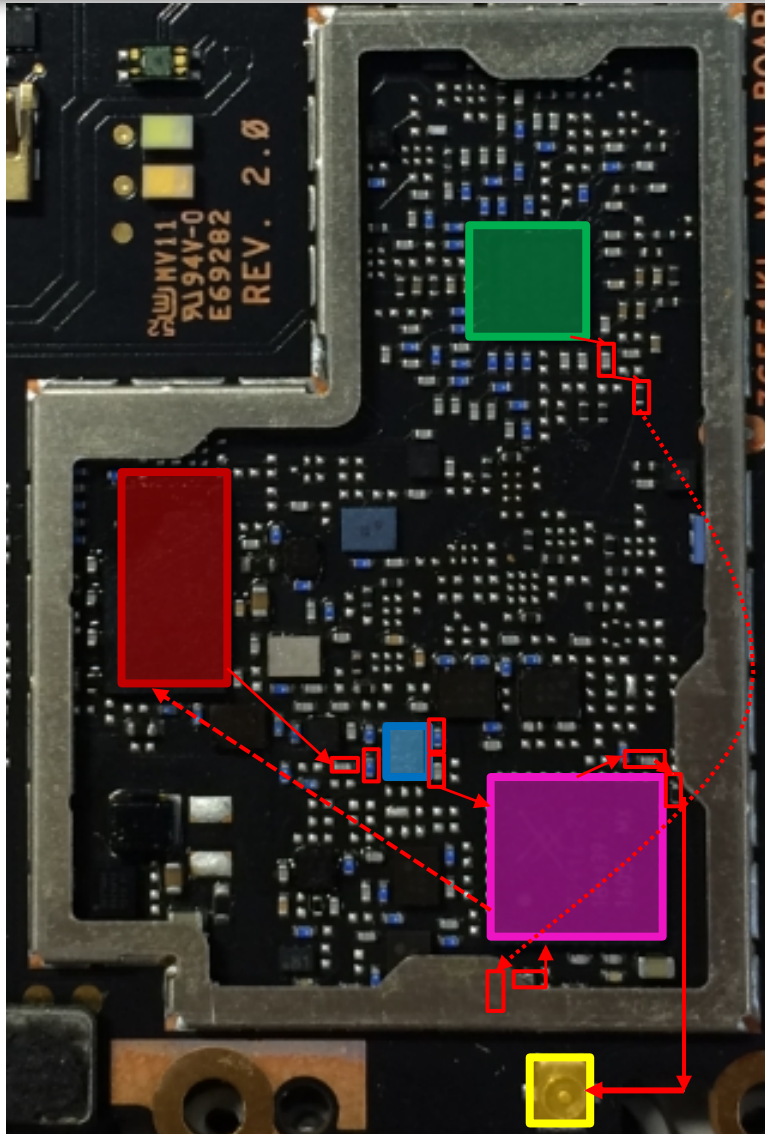
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B2 Duplexer **U4804**

TX Path : 3G&4G Band 5/6/18/19/26



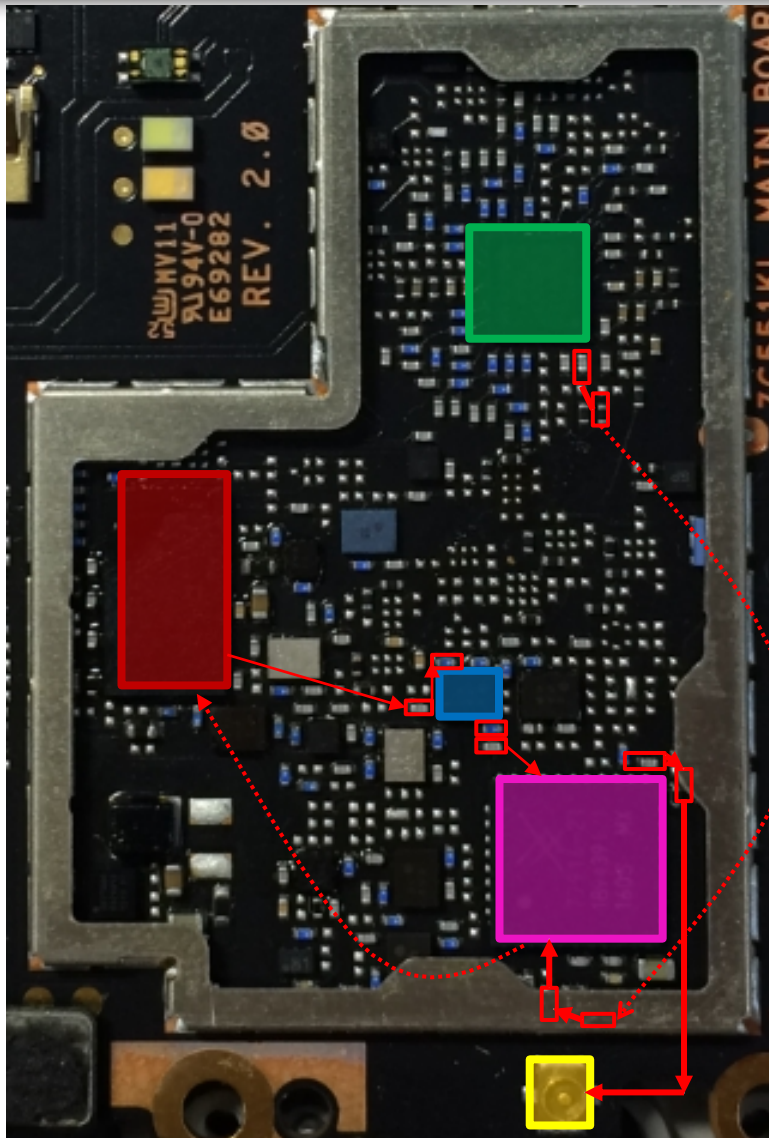
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B5/6/18/19/26 Duplexer **U4803**

TX Path : 3G&4G Band 8



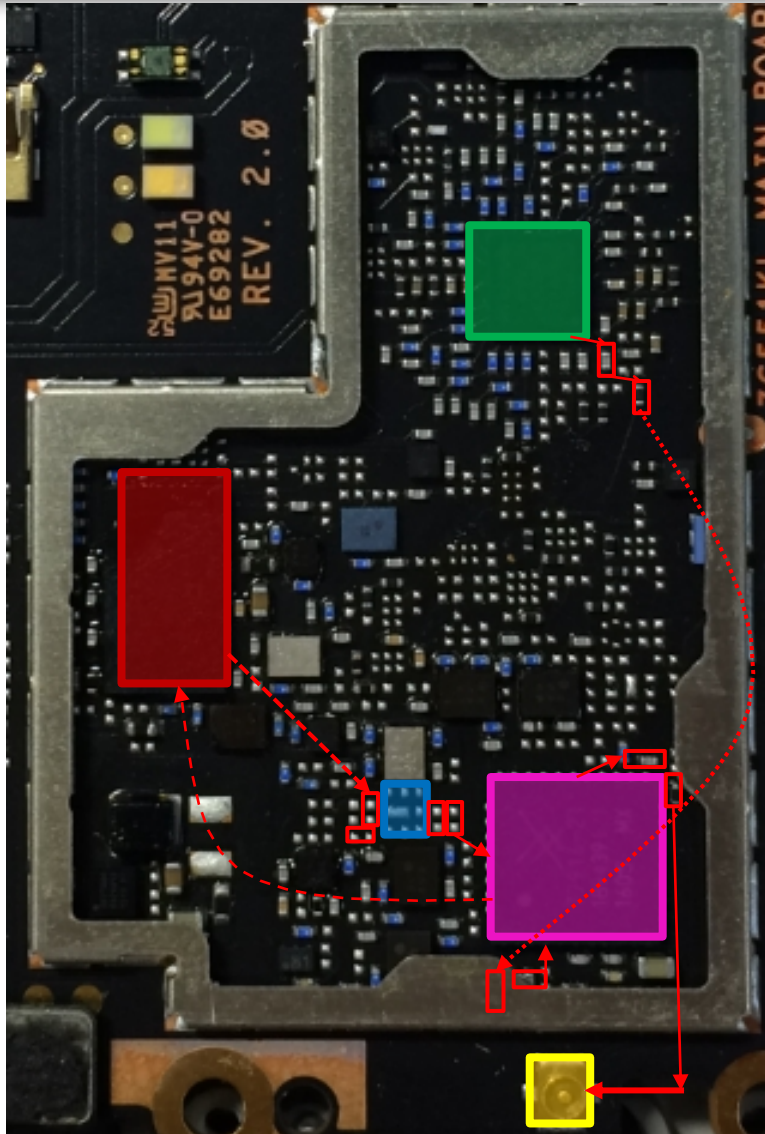
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B8 Duplexer **U4812**

TX Path : 4G Band 3



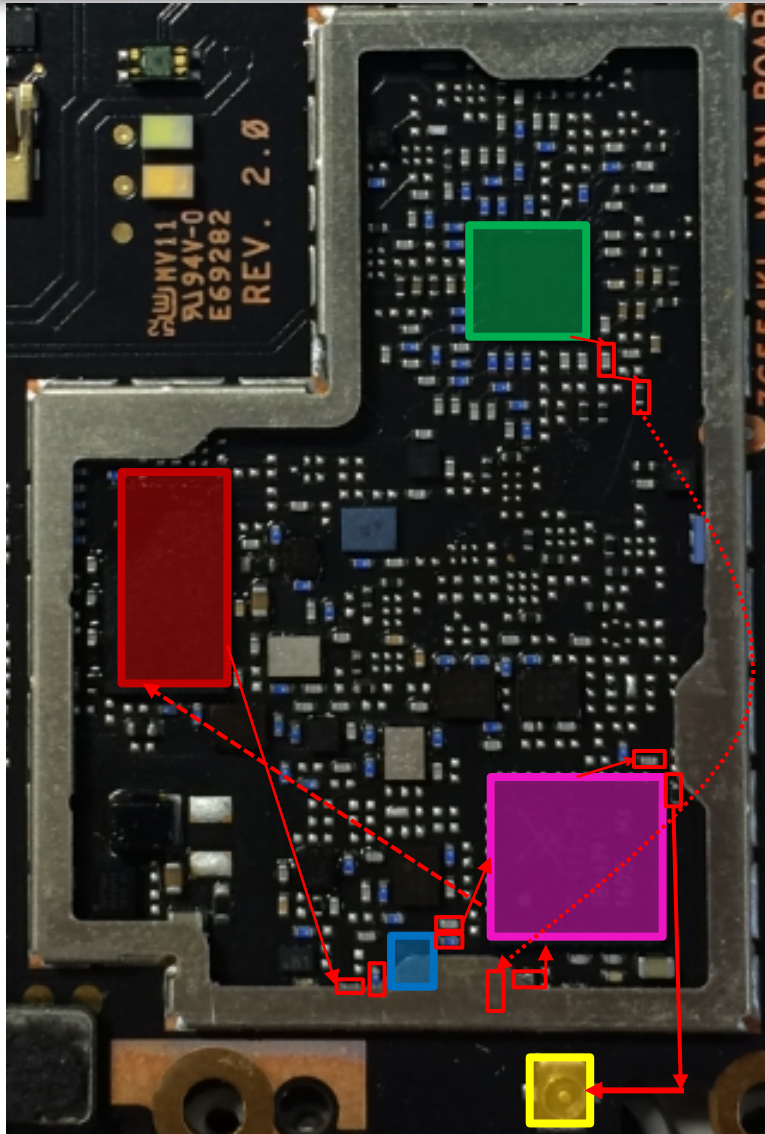
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B3 Duplexer **U4811**

TX Path : 4G Band 20



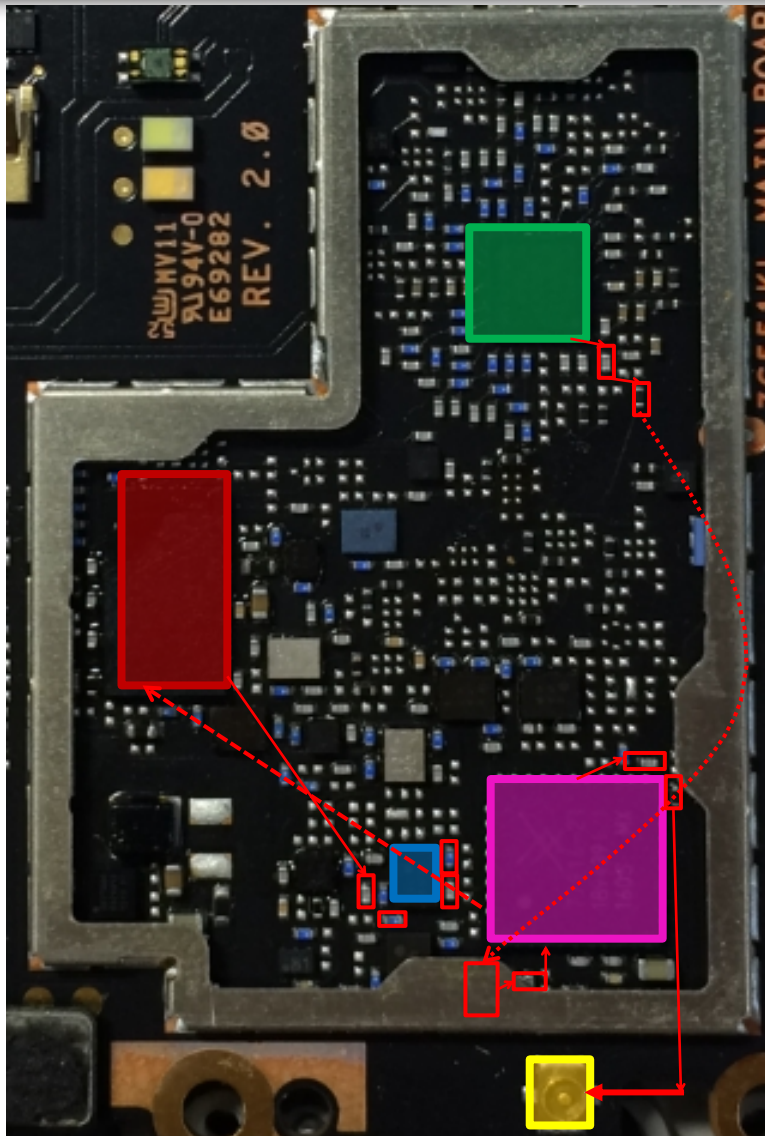
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B20 Duplexer **U4813**

TX Path : 4G Band 28A



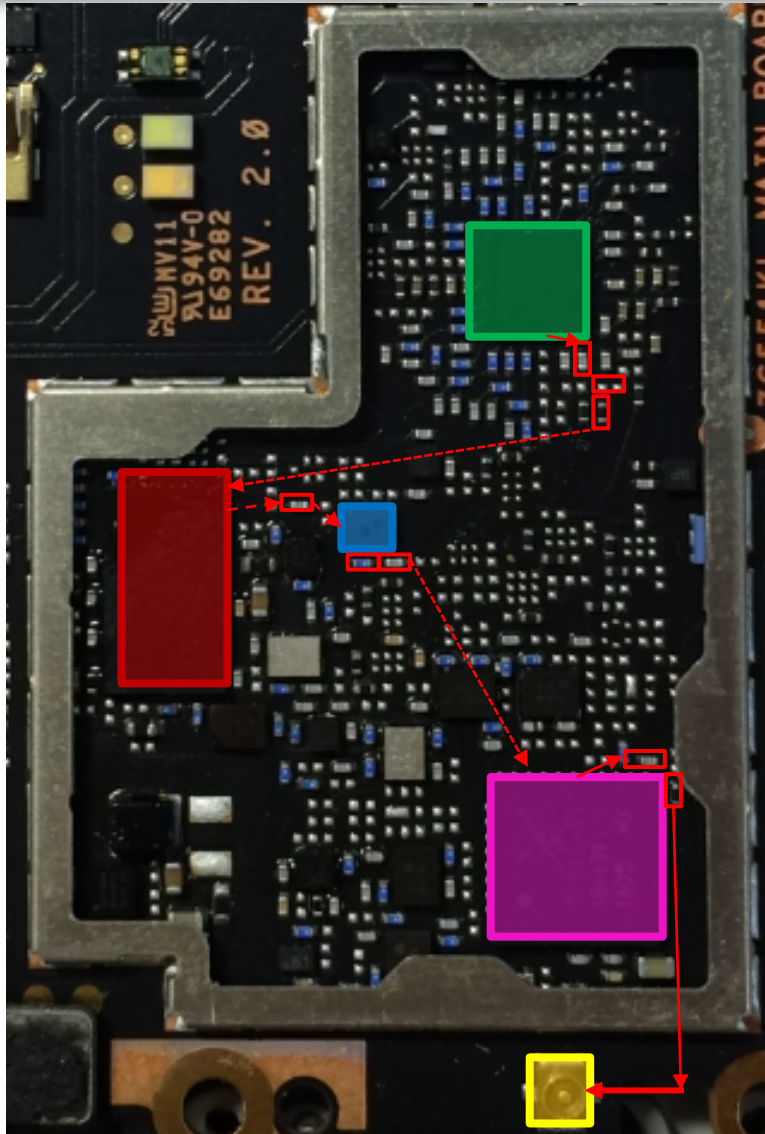
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B28B Duplexer **U4802**

TX Path : 4G Band 28B



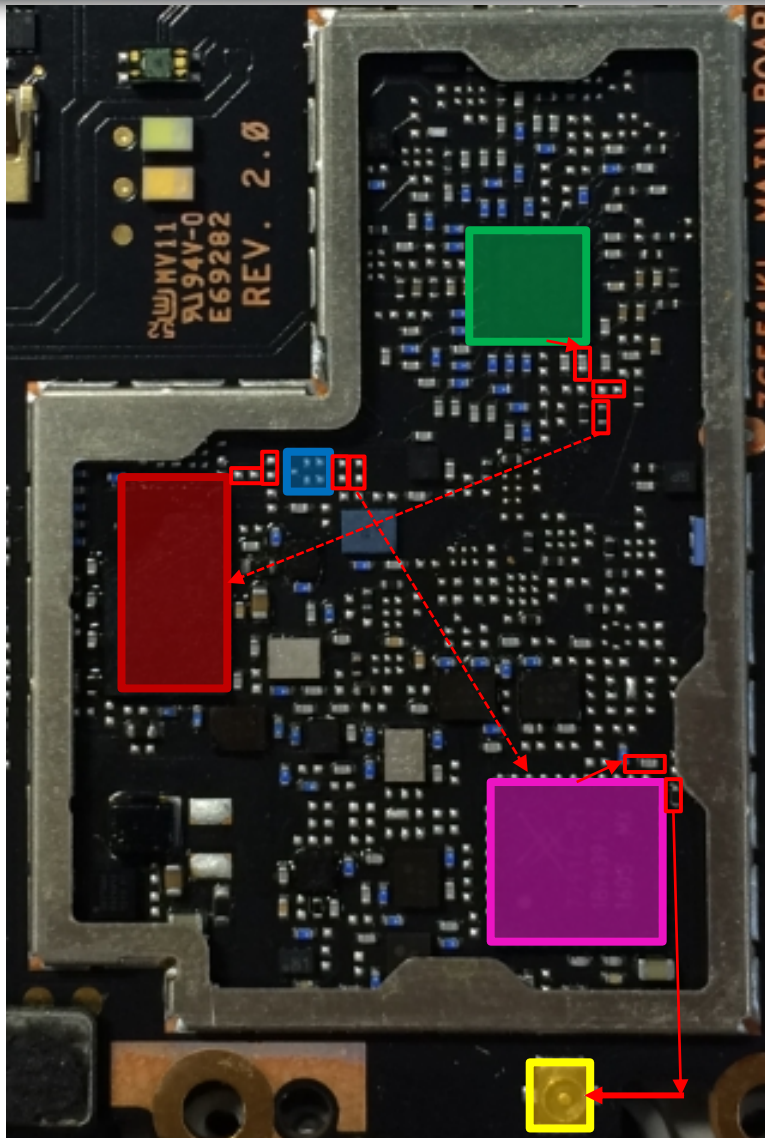
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B28B Duplexer **U4801**

TX Path : 4G Band 7



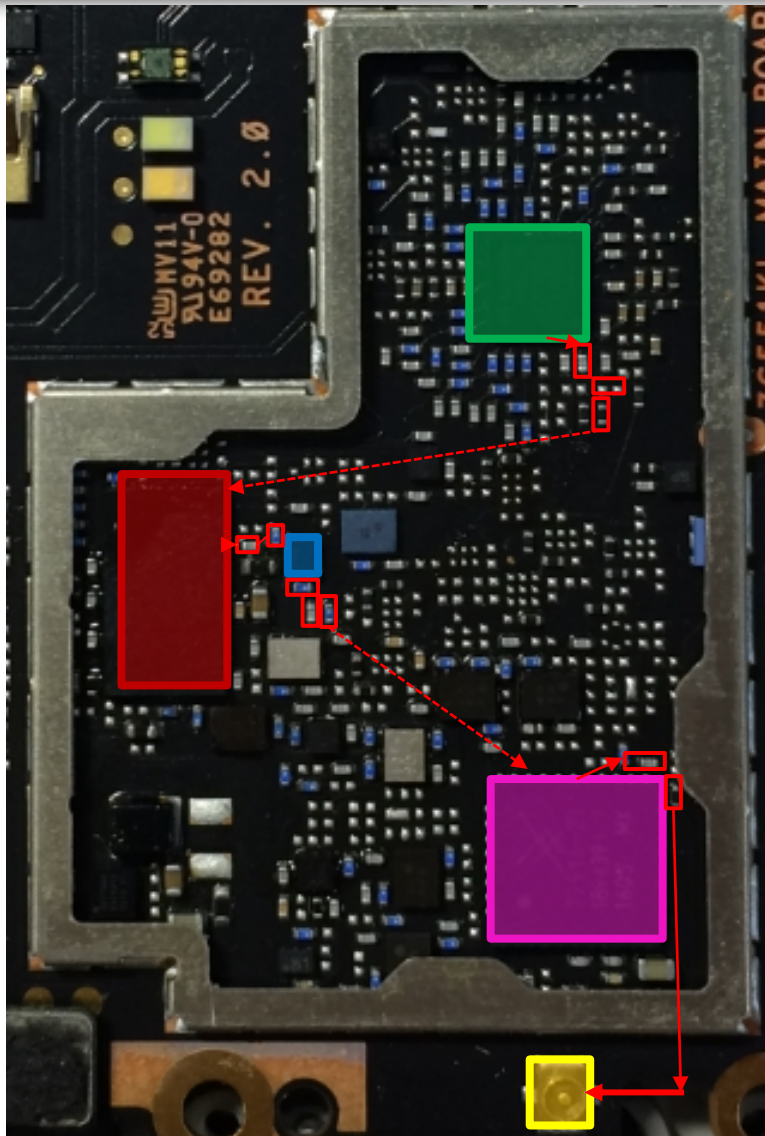
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B7 Duplexer **U4810**

TX Path : 4G Band 40



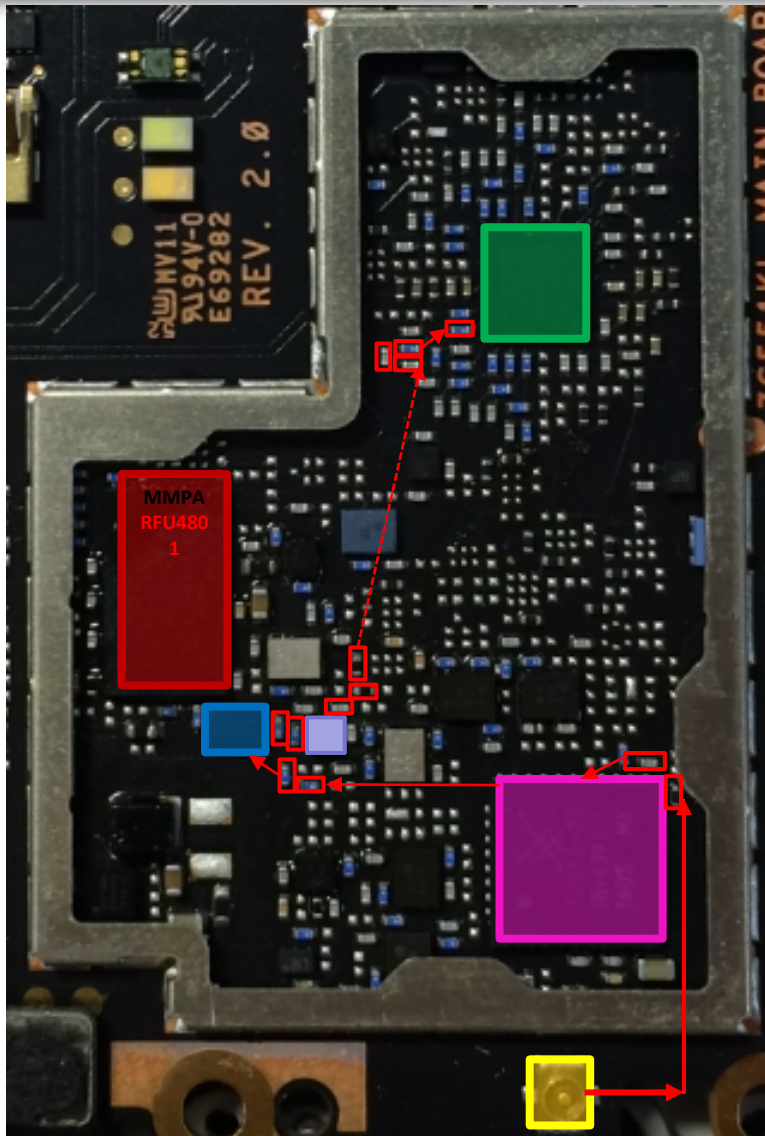
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B40 SAW Filter **U4808**

TX Path : 4G Band 38/41



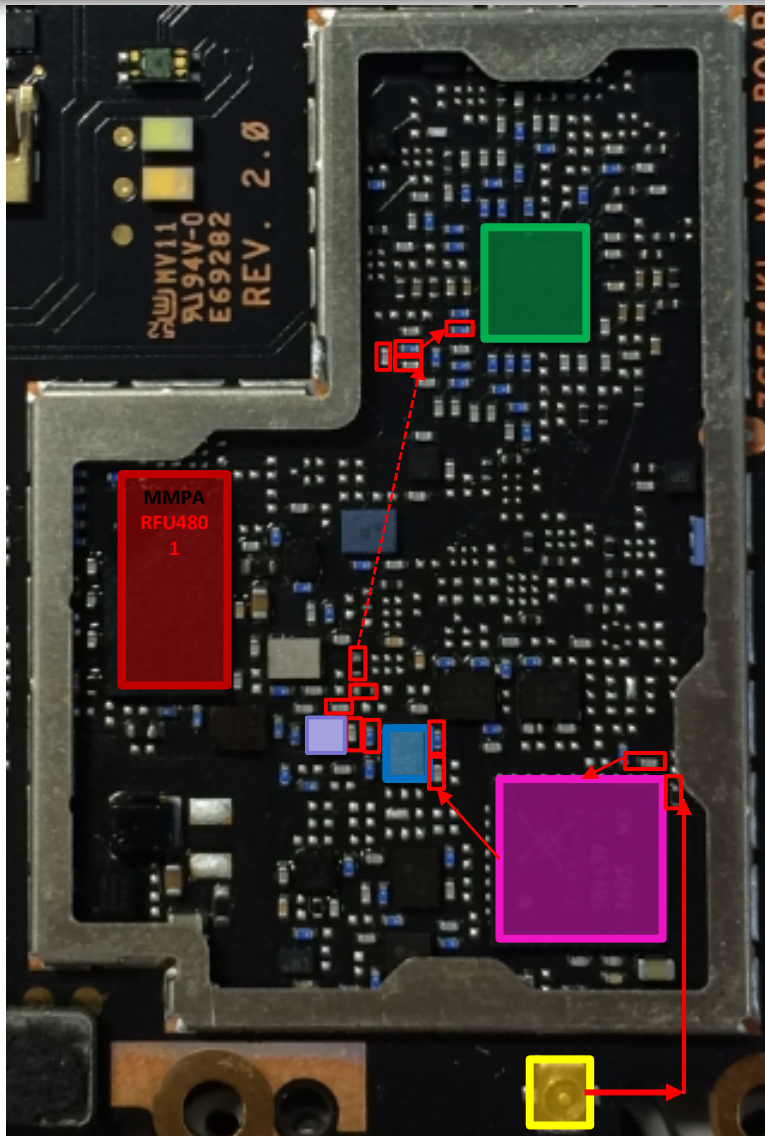
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B38/41 SAW Filter **U4809**

RX Path : 3G&4G Band5/6/17/18/19/26/GSM850



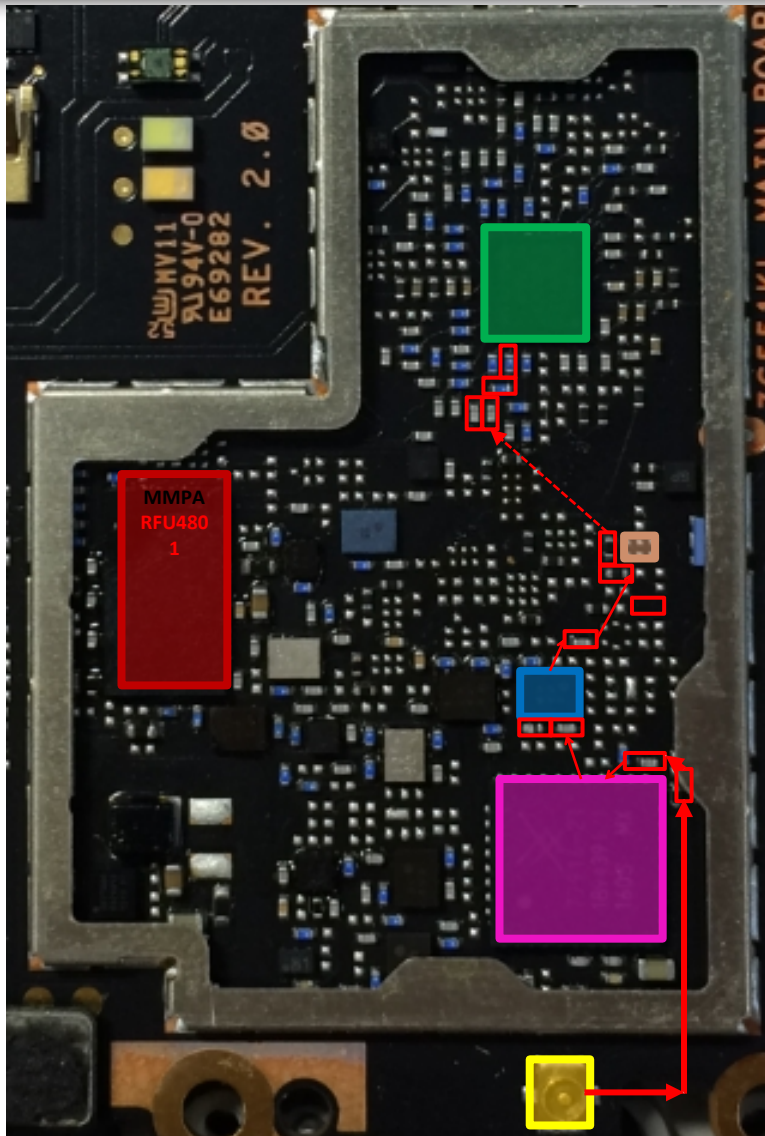
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B5/6/18/19/26 Duplexer **U4803**
-  B5/B8 SPDT **U4911**

RX Path : 3G&4G Band8/GSM900



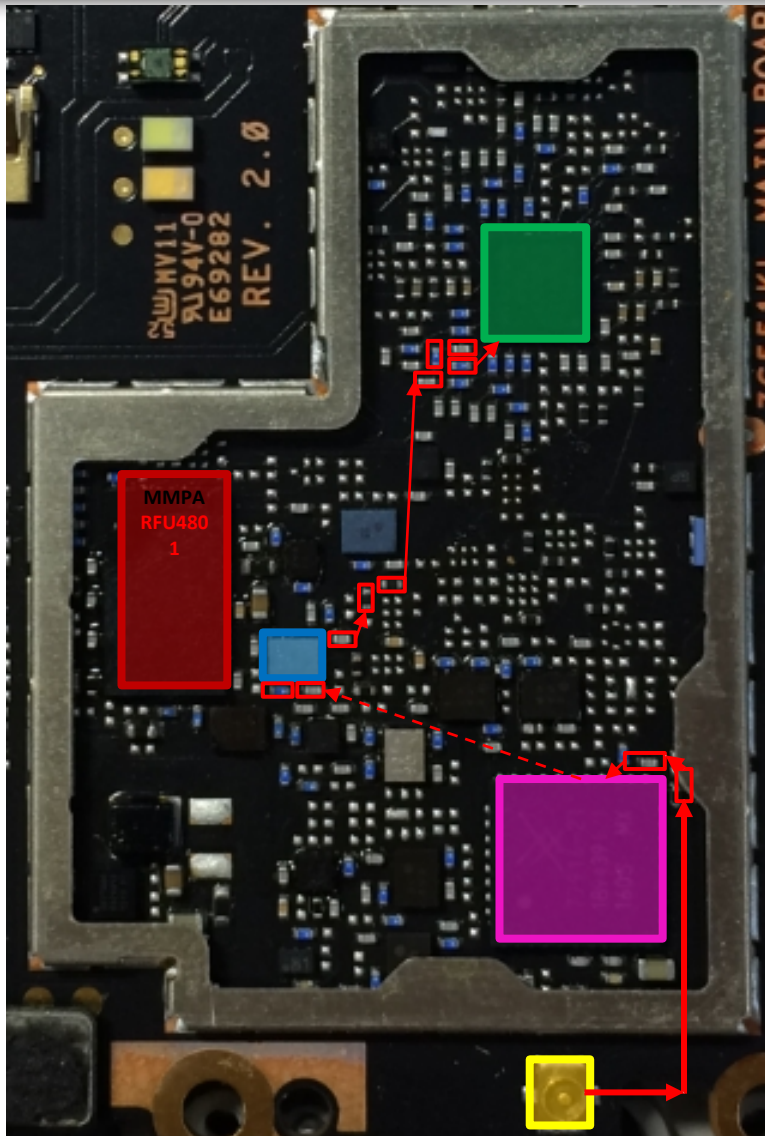
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B5/6/18/19/26 Duplexer **U4812**
-  B5/B8 SPDT **U4911**

RX Path : 3G&4G Band 1



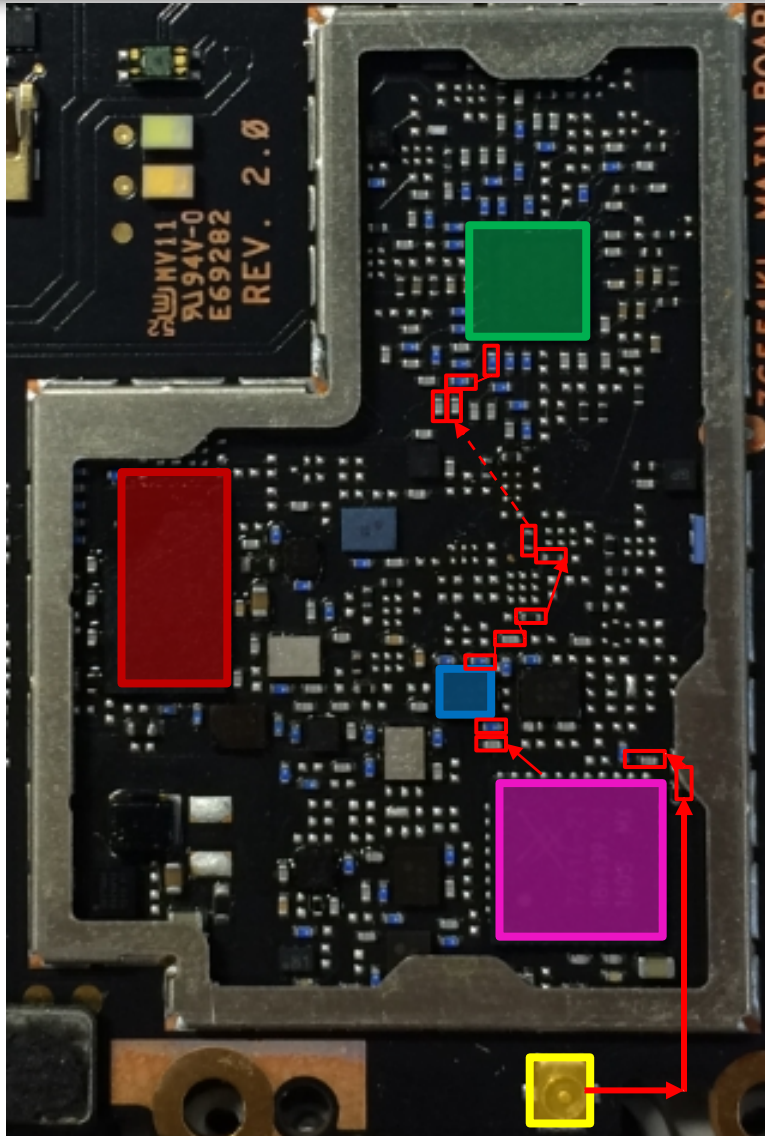
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B1 Duplexer **U4807**
-  B1/4 PRX LNA **U4903**

RX Path : 3G&4G Band 2/PCS



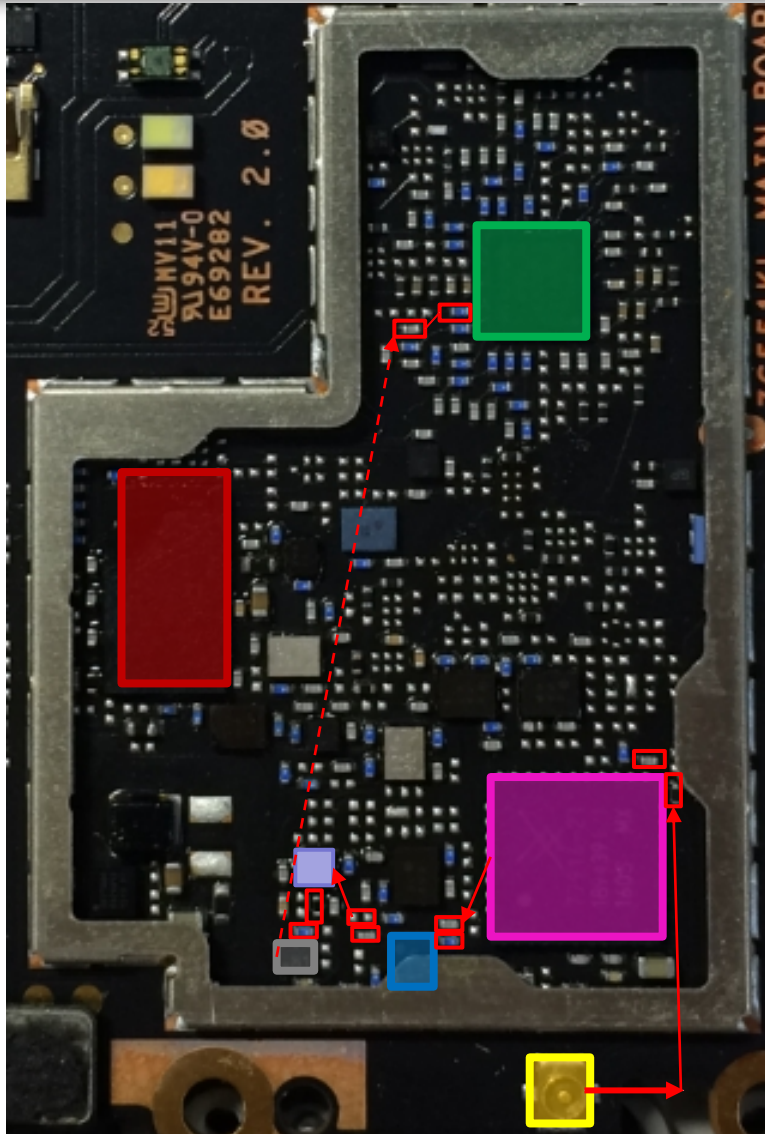
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B2 Duplexer **U4804**

RX Path : 4G Band 3/DCS



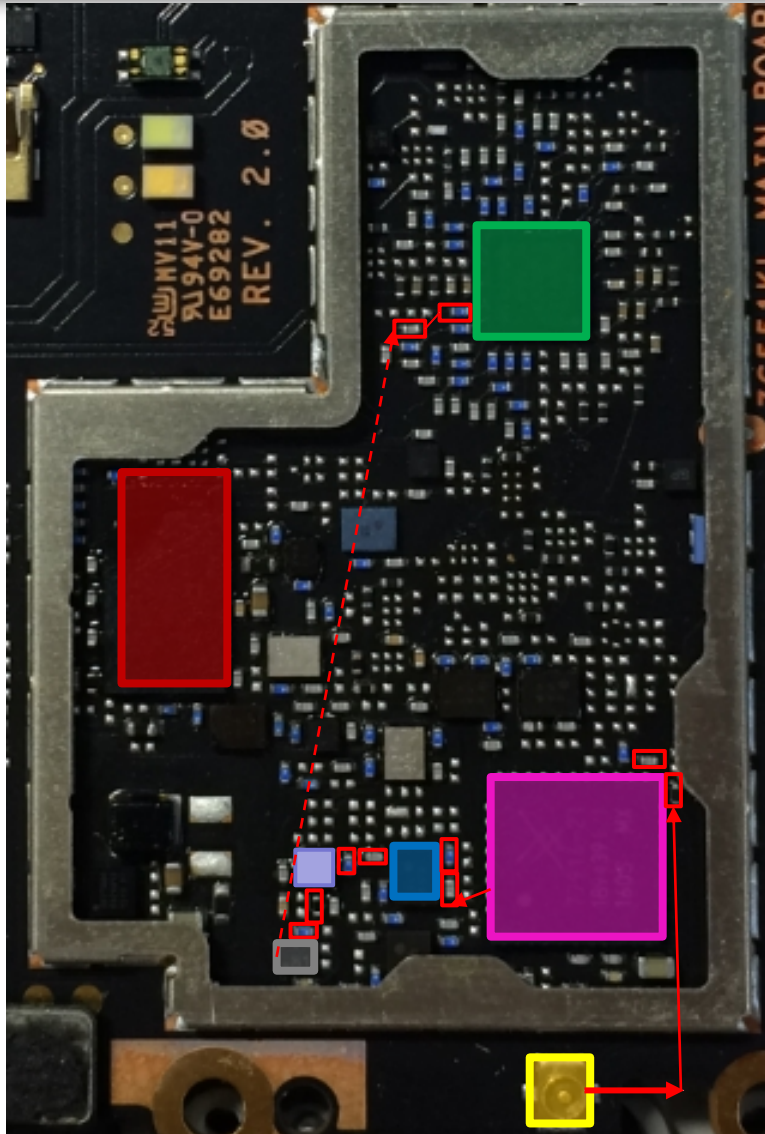
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B3 Duplexer **U4811**

RX Path : 4G Band 28A



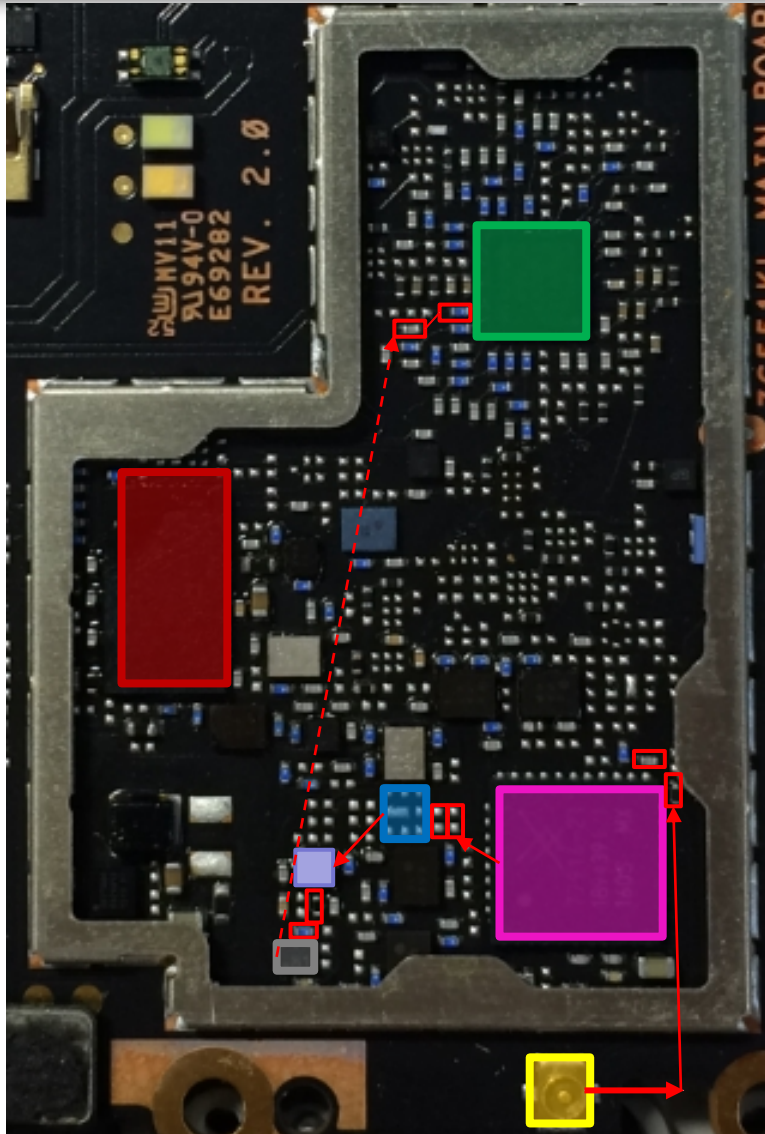
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B28B Duplexer **U4802**
-  B28A/B28B/B20 PRX LNA **U4907**
-  B28A/B28B/B20 SP3T Switch **U4902**

RX Path : 4G Band 28B



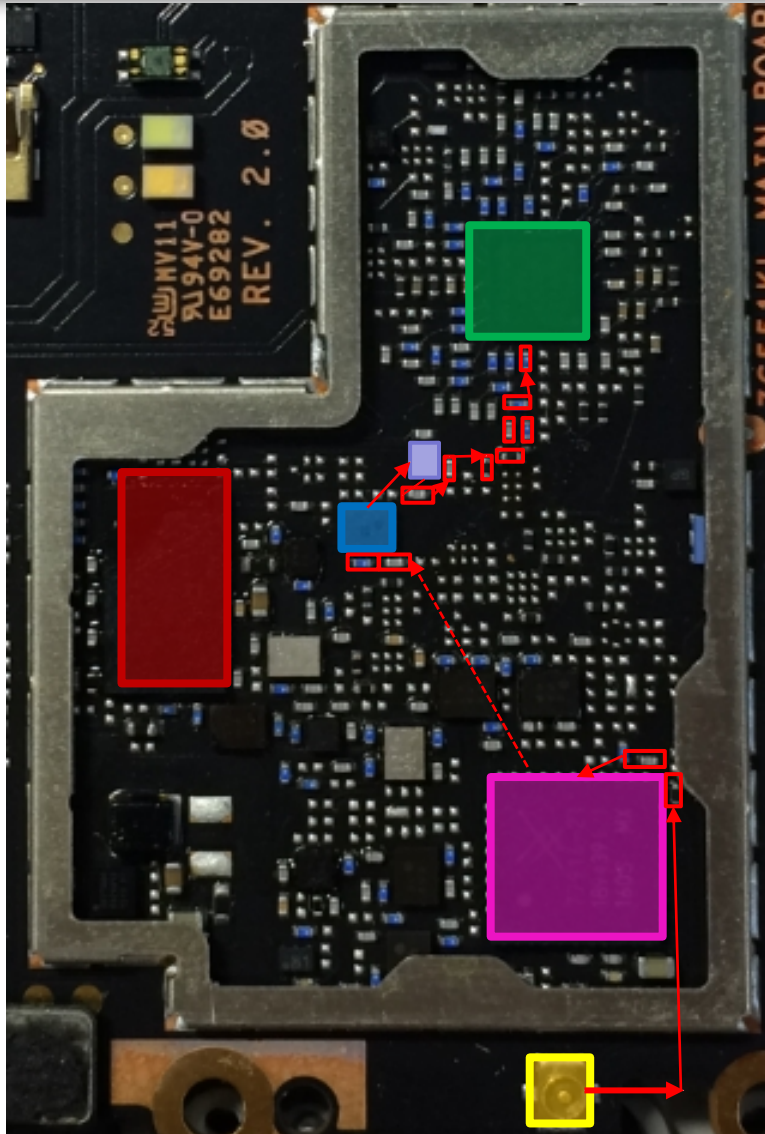
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B28B Duplexer **U4801**
-  B28A/B28B/B20 PRX LNA **U4907**
-  B28A/B28B/B20 SP3T Switch **U4902**

RX Path : 4G Band 20



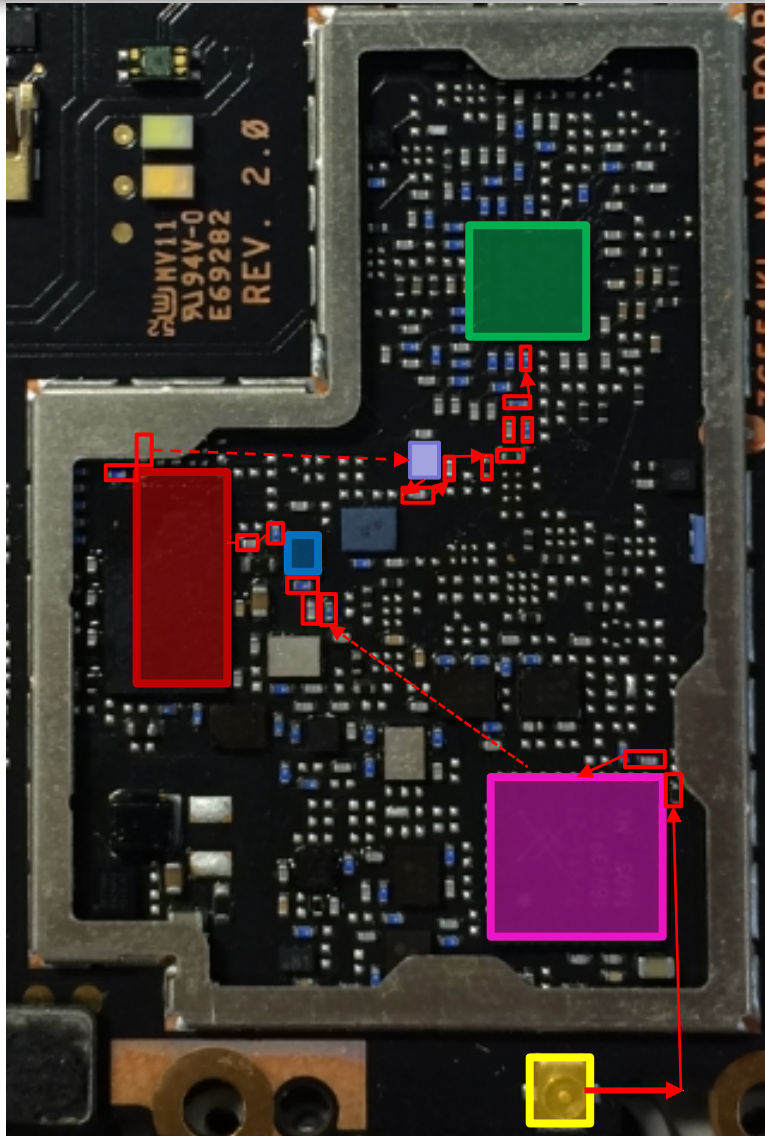
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B28B Duplexer **U4807**
-  B28A/B28B/B20 PRX LNA **U4913**
-  B28A/B28B/B20 SP3T Switch **U4902**

RX Path : 4G Band 7



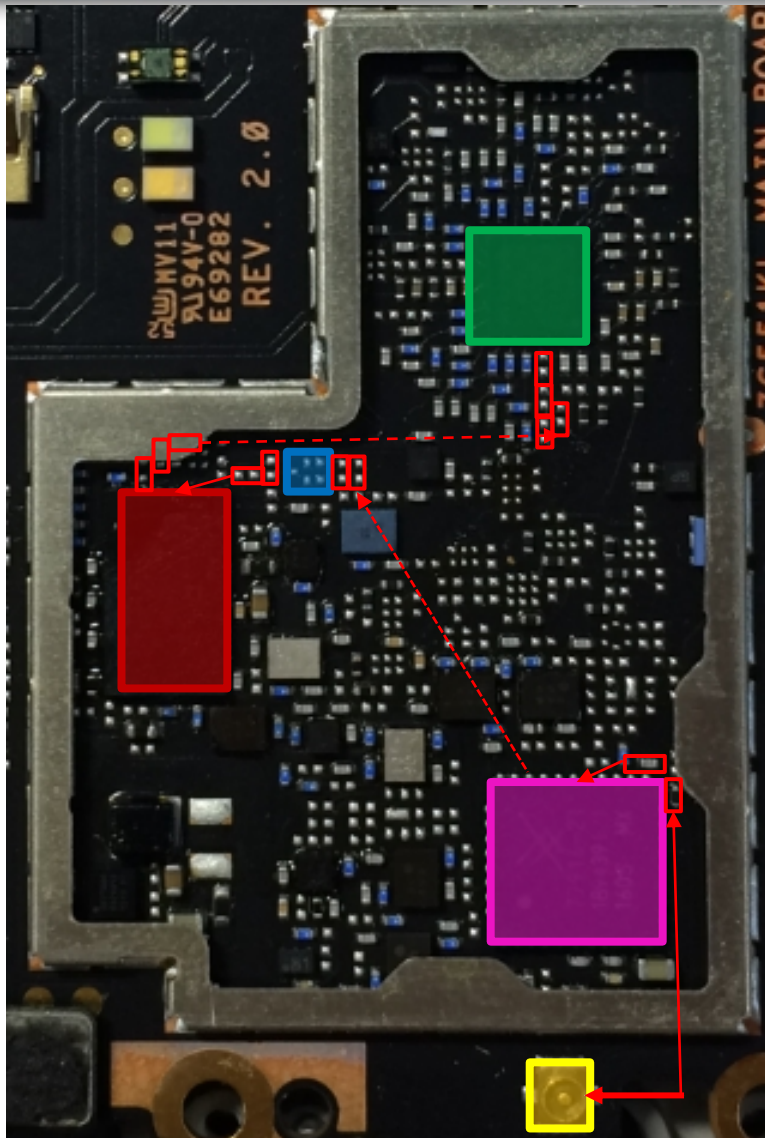
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B7 Duplexer **U4810**
-  B7/B38/B41 SPDT Switch **U4912**

RX Path : 4G Band 38/41



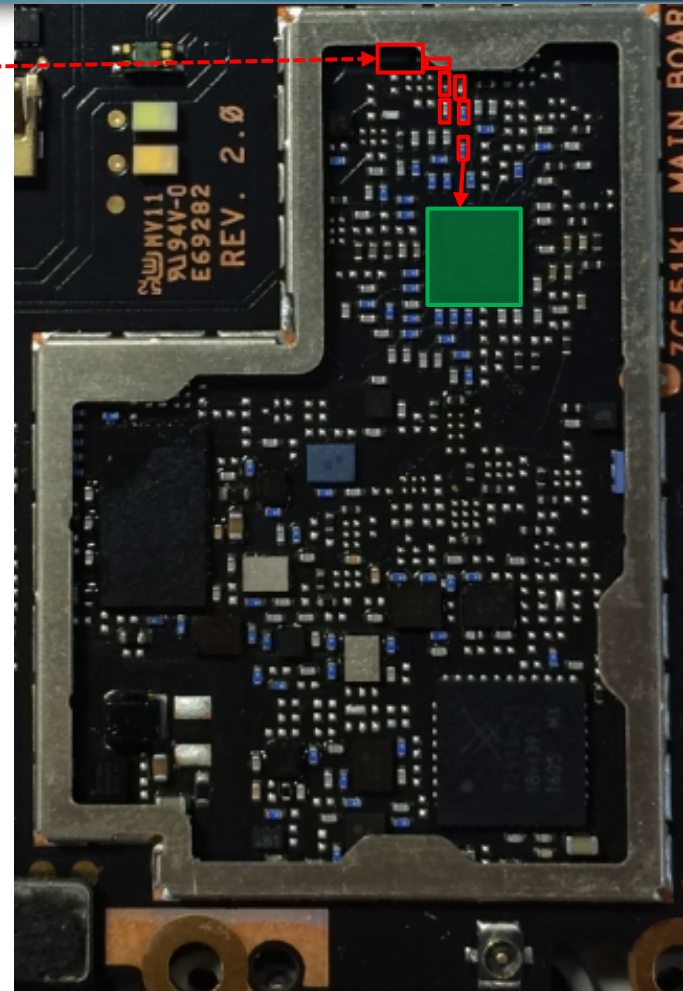
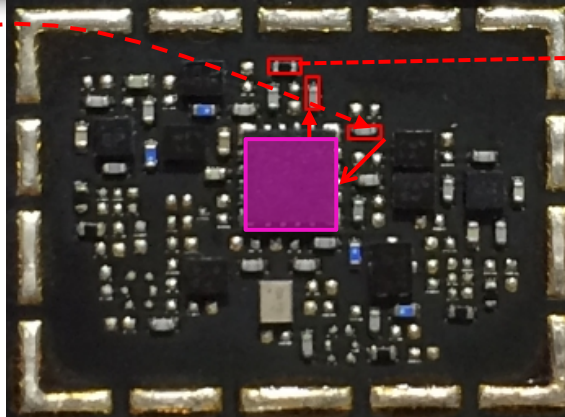
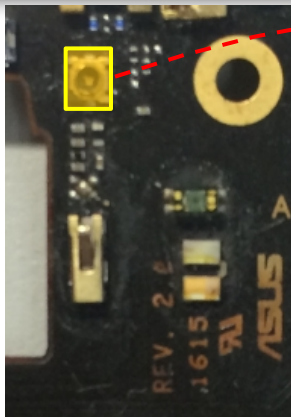
-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B38/B41 SAW Filter **U4809**
-  B7/B38/B41 SPDT **U4912**

RX Path : 4G Band 40



-  Main Antenna SW **U4701**
-  MMPA **U4805**
-  RF Main Connector **CON4701**
-  Transceiver **U4501**
-  B40 SAW Filter **U4808**

Diversity RX Path : 3G/4G Band 1; 4G Band 4



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**

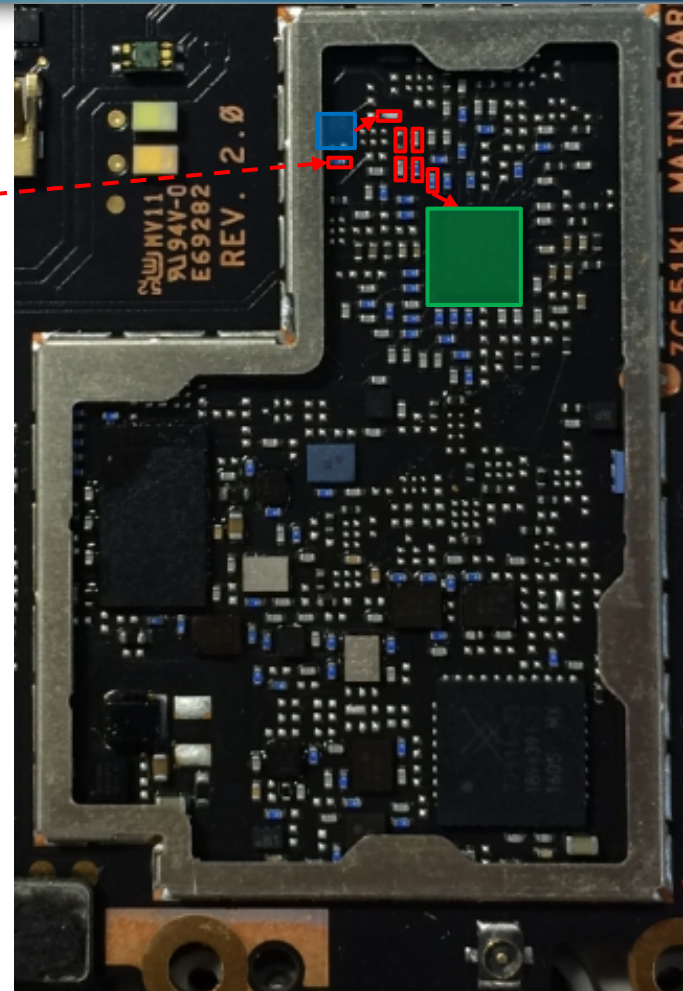
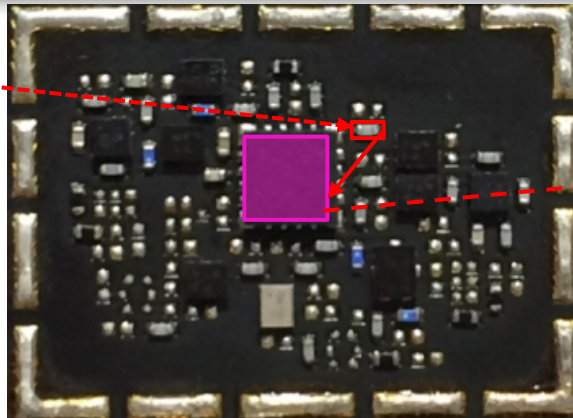
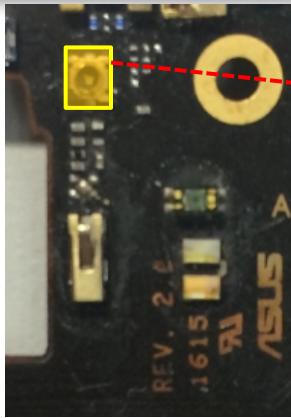


Transceiver **U4501**



B1 SAW Filter **U5105**

Diversity RX Path : 3G/4G Band 2



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**

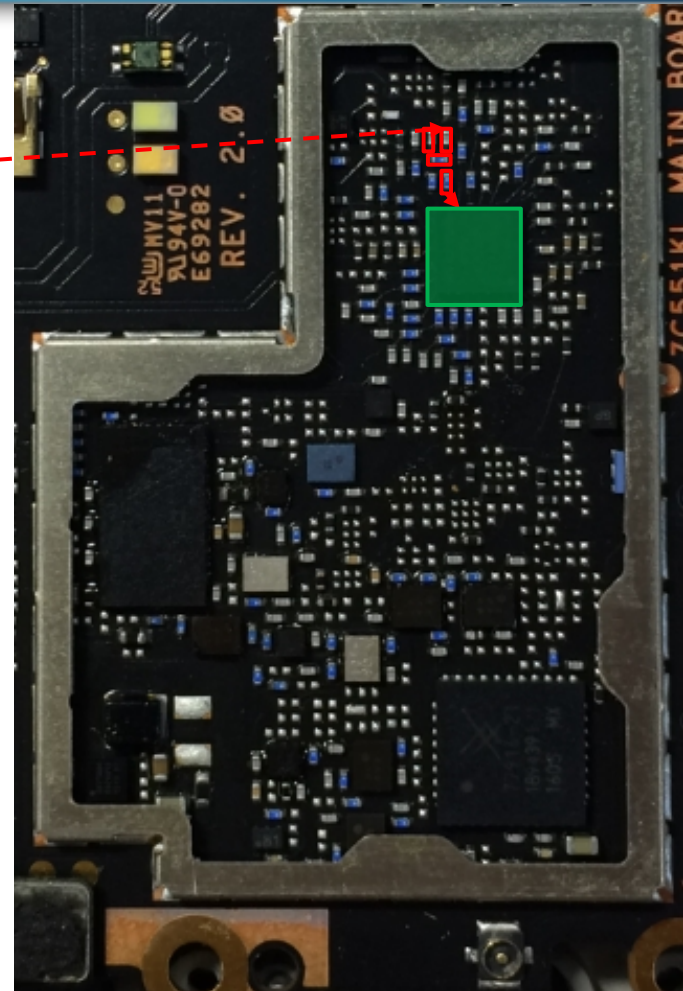
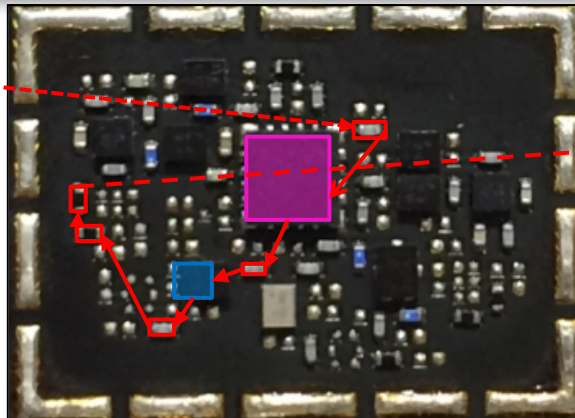
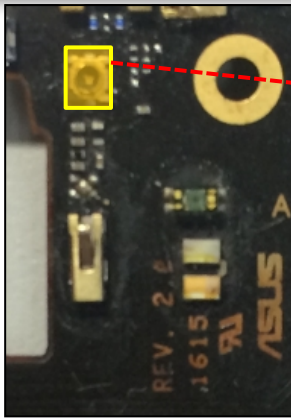


Transceiver **U4501**



B2 SAW Filter **U5110**

Diversity RX Path : 4G Band 3



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**

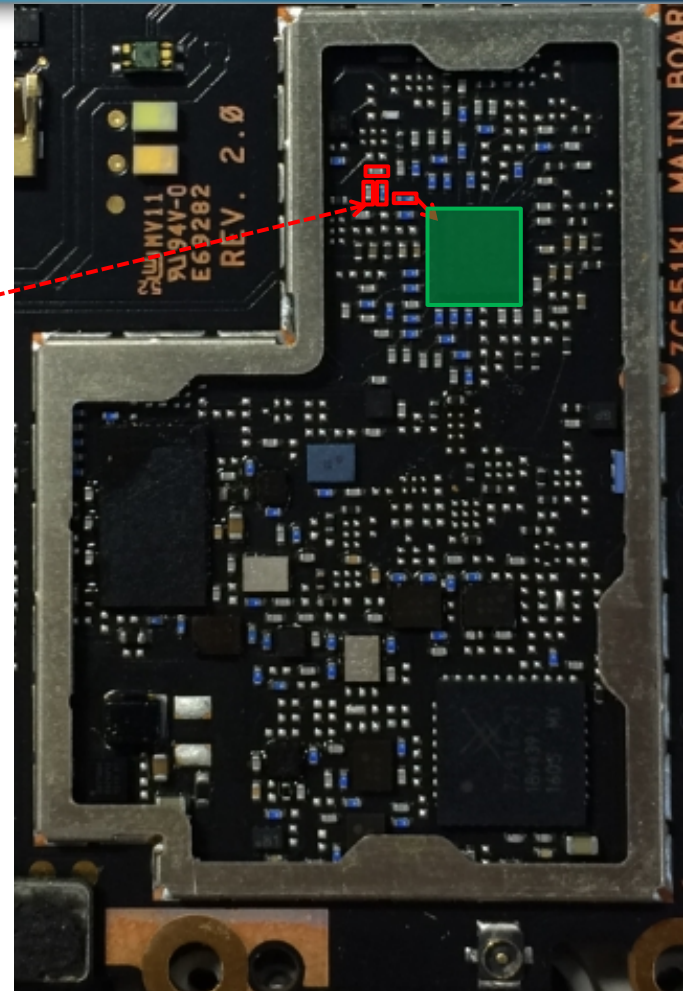
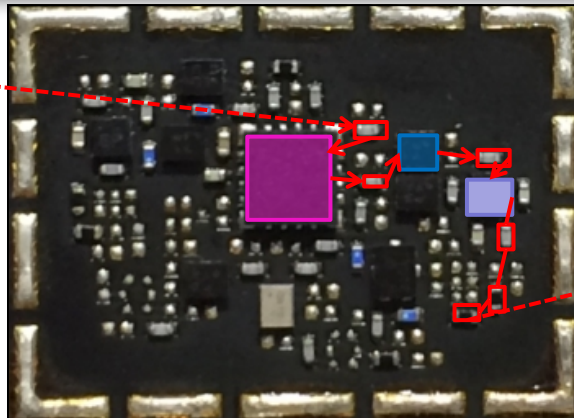
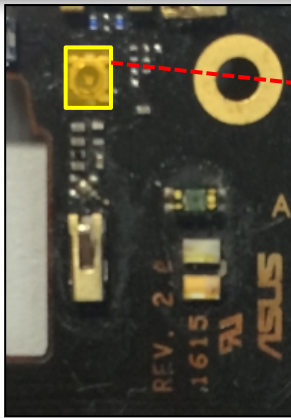


Transceiver **U4501**



B3 SAW Filter **U5107**

Diversity RX Path : 3G/4G Band 5



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**



Transceiver **U4501**

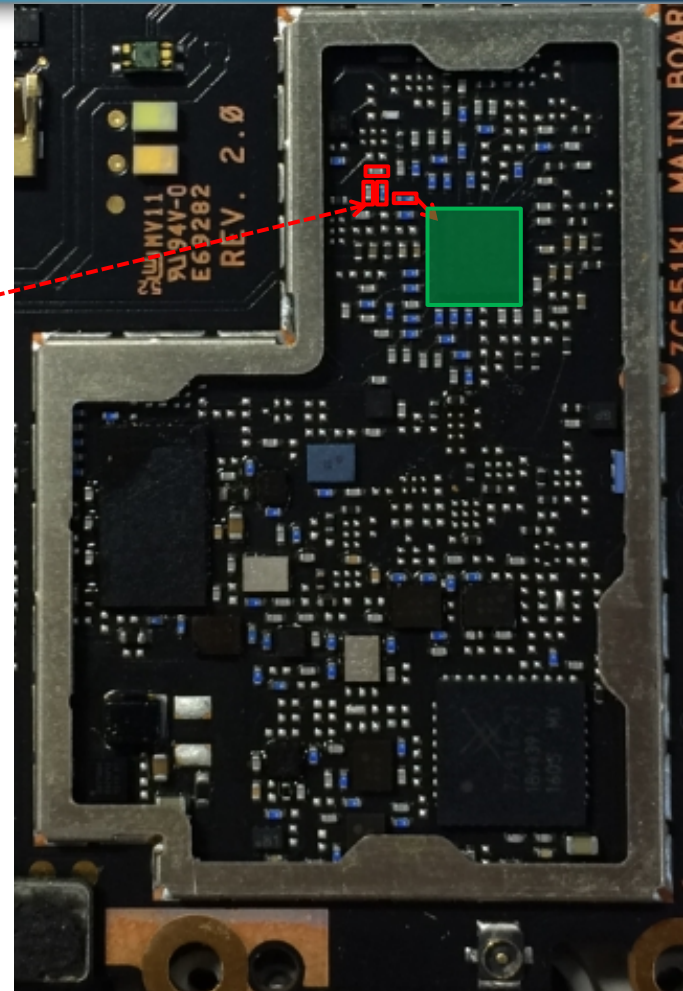
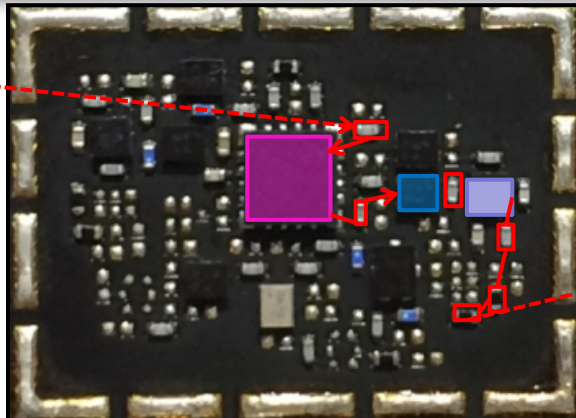
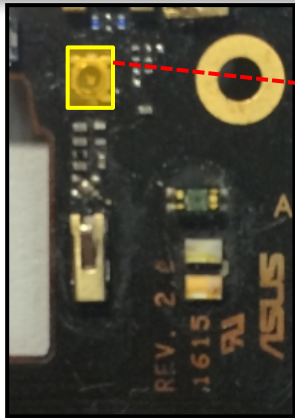


B5 SAW Filter **U5121**



B5/B8 SPDT Switch **U5102**

Diversity RX Path : 3G/4G Band 8



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**



Transceiver **U4501**

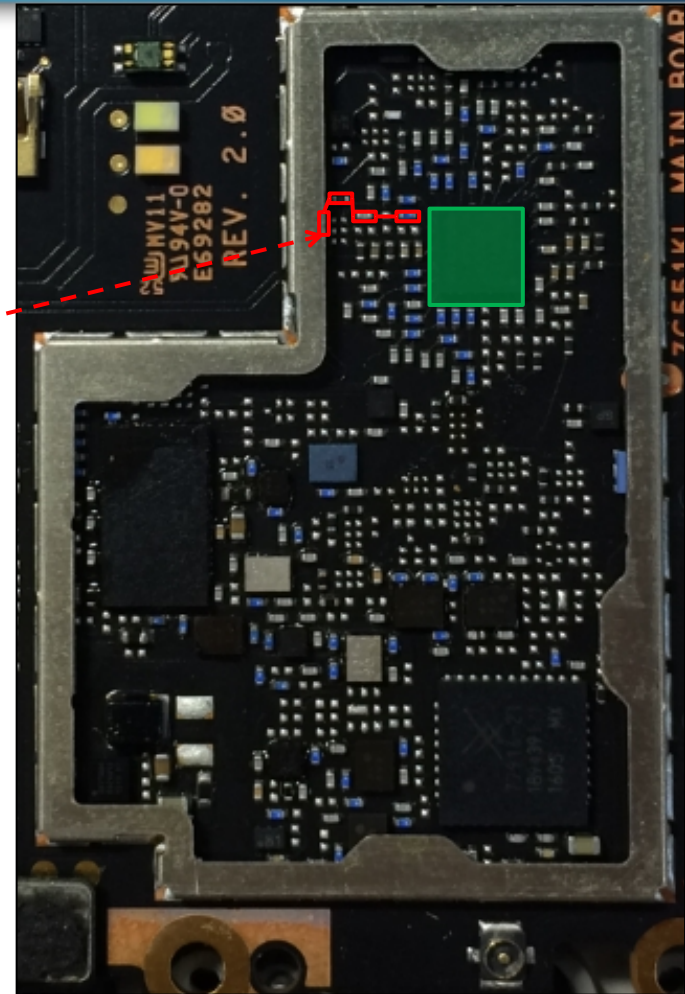
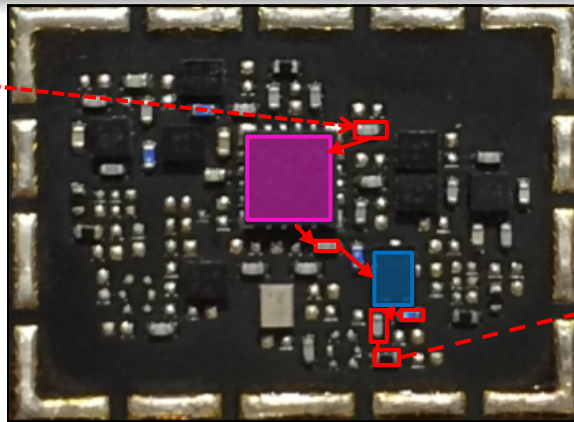
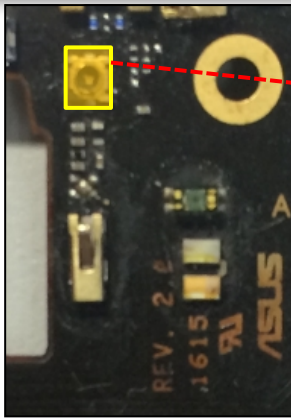


B8 SAW Filter **U5118**



B5/B8 SPDT Switch **U5102**

Diversity RX Path : 4G Band 28/Band20



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**

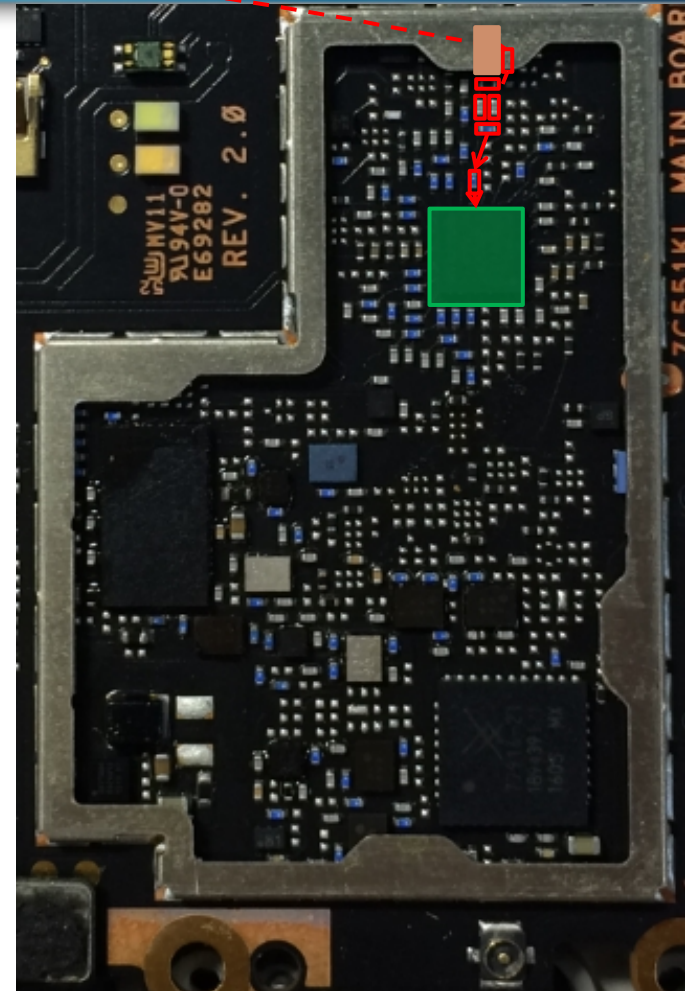
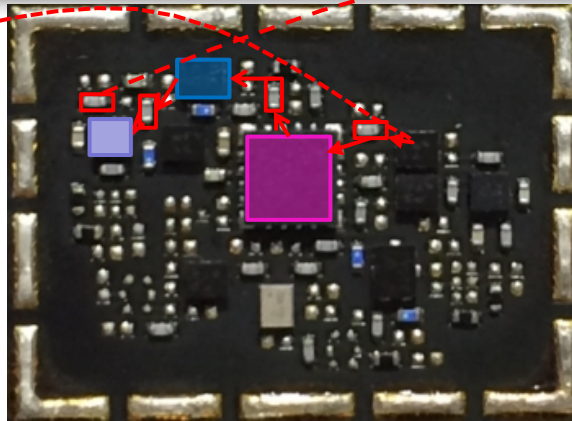
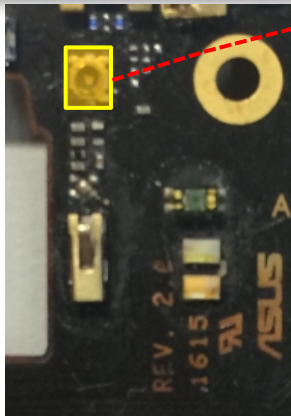


Transceiver **U4501**



B28 or B20 SAW Filter **U5114**

Diversity RX Path : 4G Band 7



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**



Transceiver **U4501**



B7 SAW Filter **U5109**

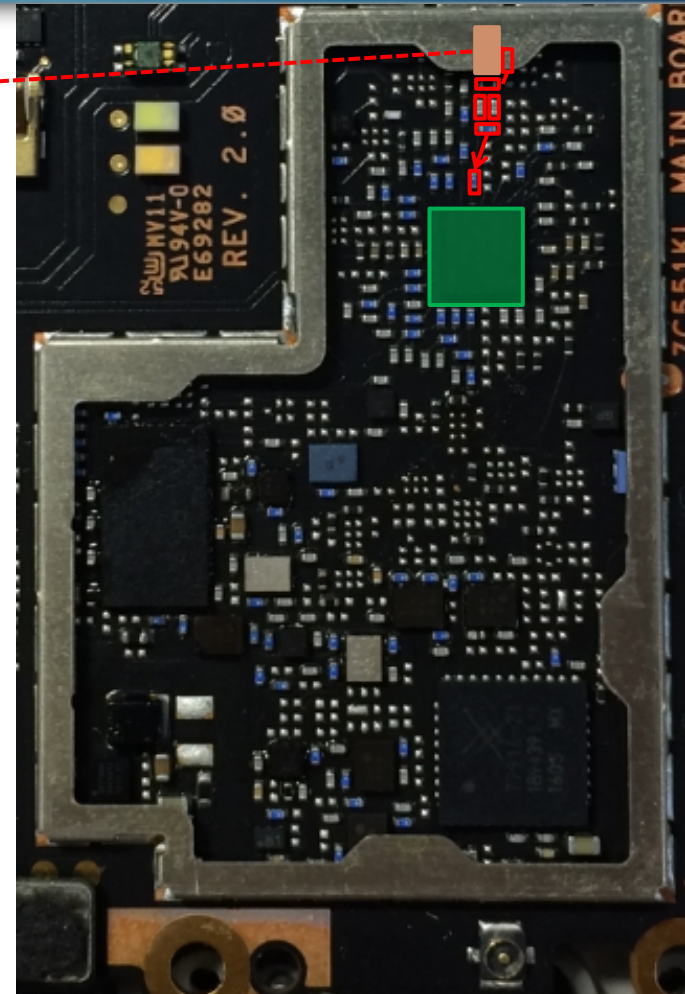
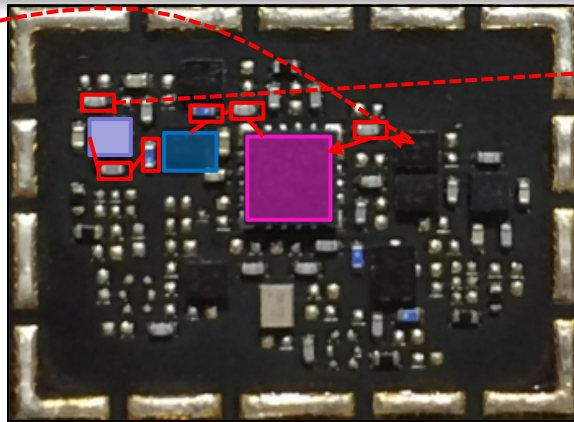
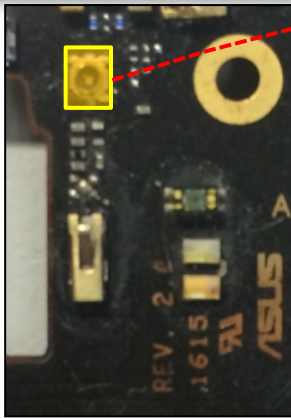


B7/B38/B41 SPDT **U5122**



B7/B38/B41 DRX LNA **U5111**

Diversity RX Path : 4G Band 38/41



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**



Transceiver **U4501**



B7 SAW Filter **U5120**

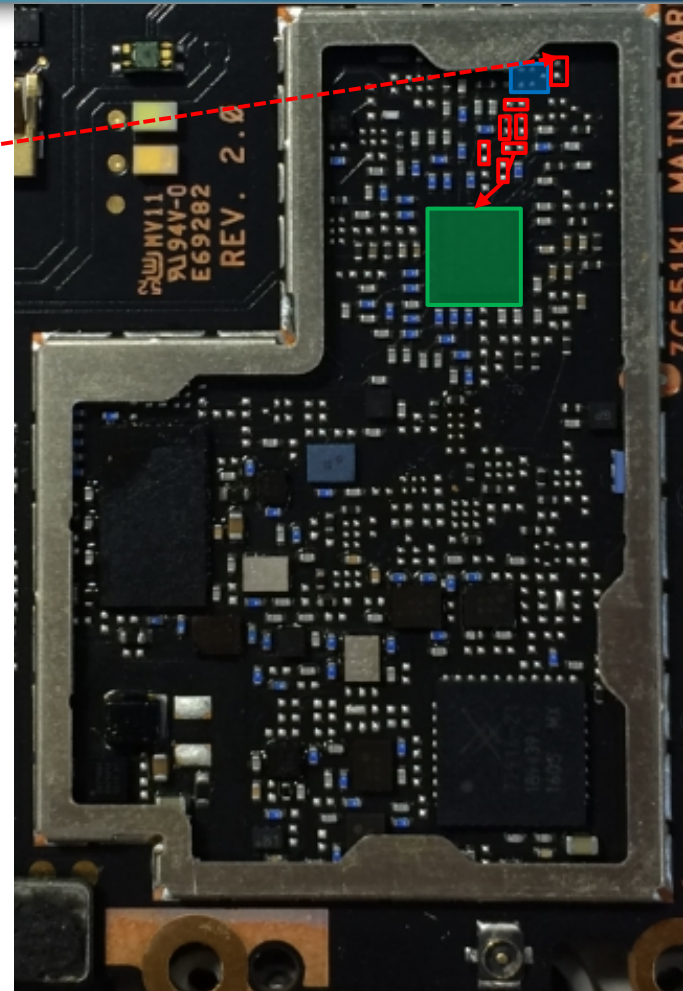
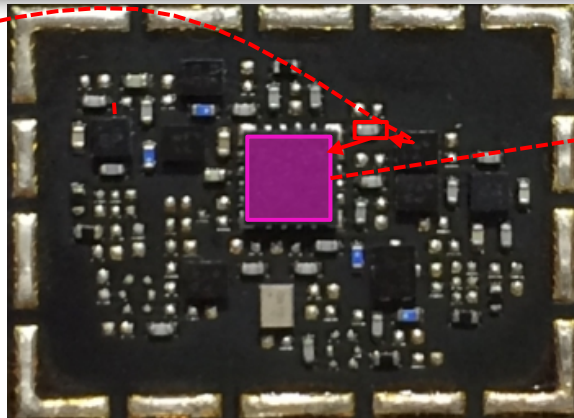
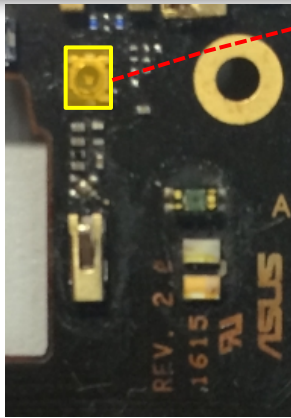


B7/B38/B41 SPDT **U5122**



B7/B38/B41 DRX LNA **U5111**

Diversity RX Path : 4G Band40



Diversity Antenna SW **U4705**



RF Diversity Connector **CON4702**

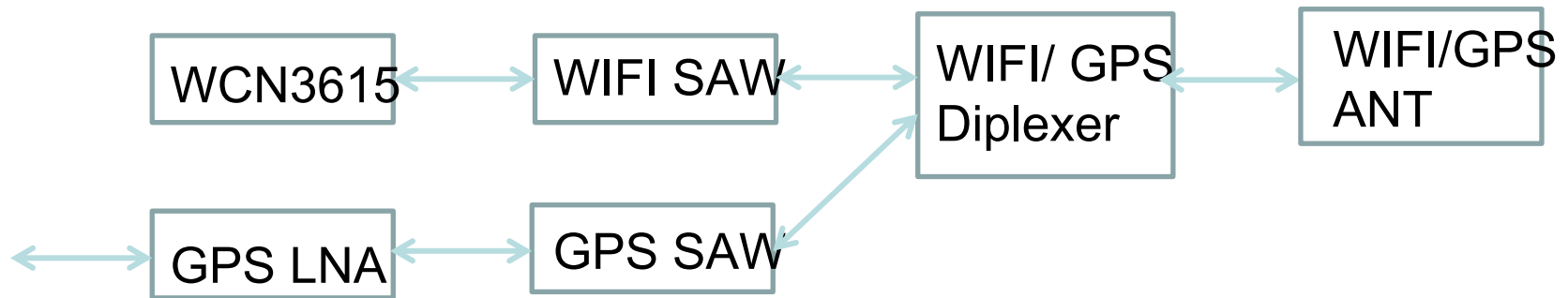


Transceiver **U4501**



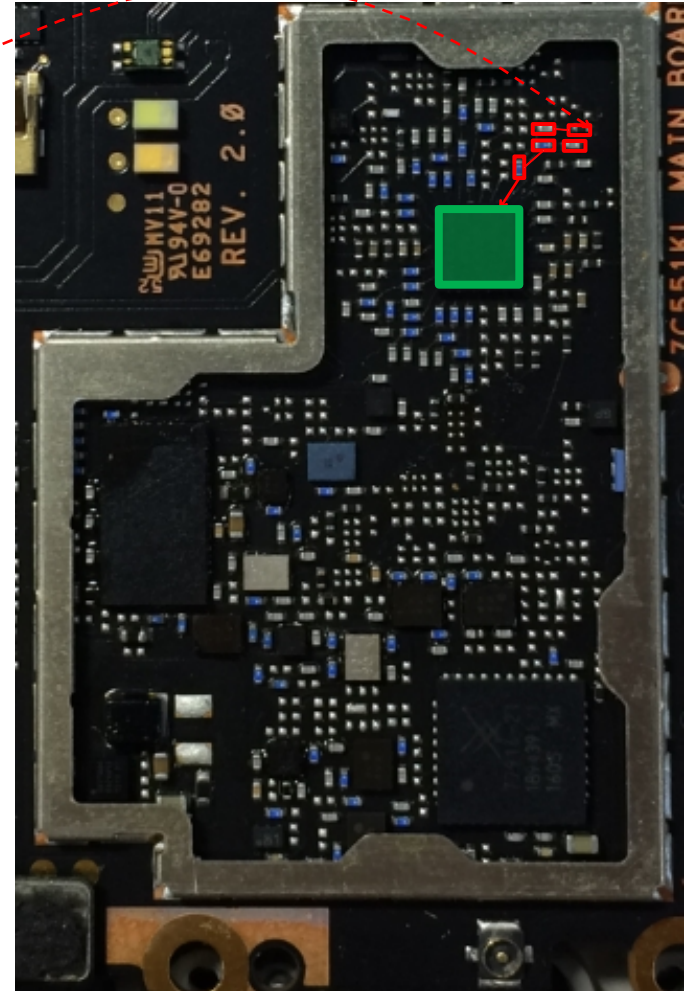
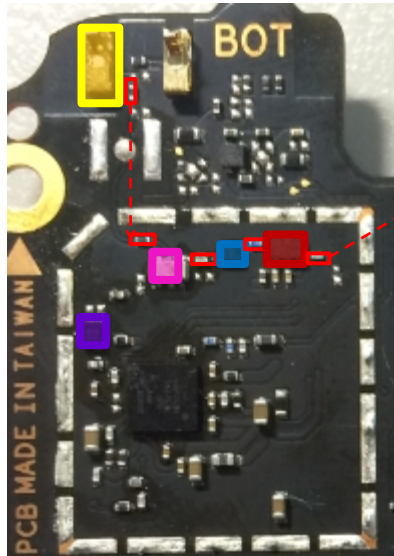
B40 SAW Filter **U5108**

WIFI & GPS Block Diagram



Back To
WTR 2956
transceiver

GPS Path



Transceiver **U4501**



GPS/Wifi Spring **ANT5201**



GPS/Wifi Diplexer **U5202**



GPS SAW Filter **U5302**



GPS LNA **U5301**

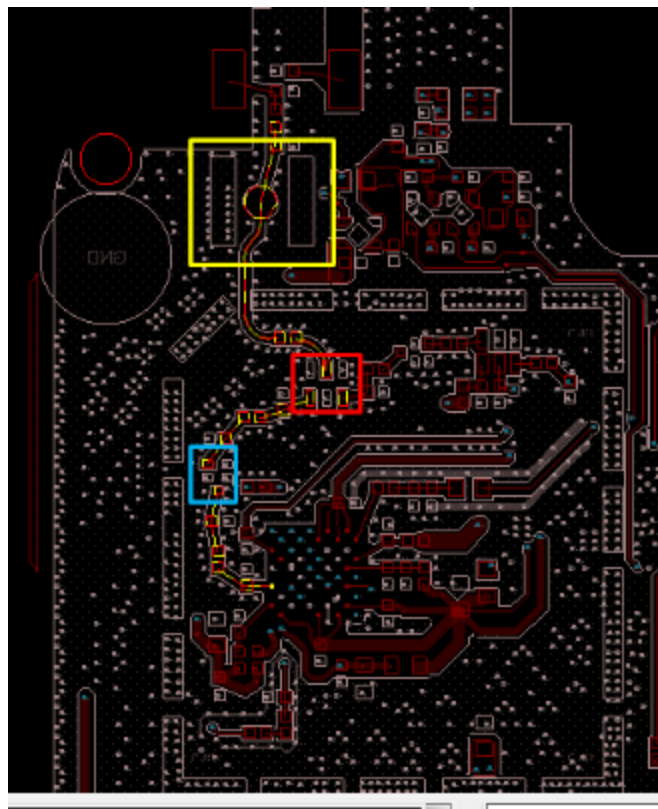


WIFI SAW FILTER **U5203**

GPS Function Test Procedure

- Check GPS function with GPS Test Tool.
- Check the GPS spring.
- Execute the following commands in the file folder with adb.exe file to monitor the real-time GPS C/N
adb shell /data/data/GpsTest 4 70
- Check U5301 (LNA) and surrounding components

BT/WiFi path



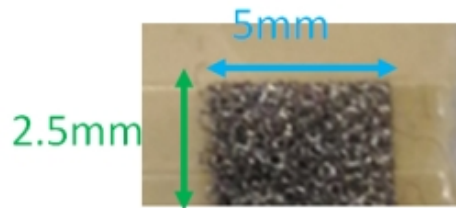
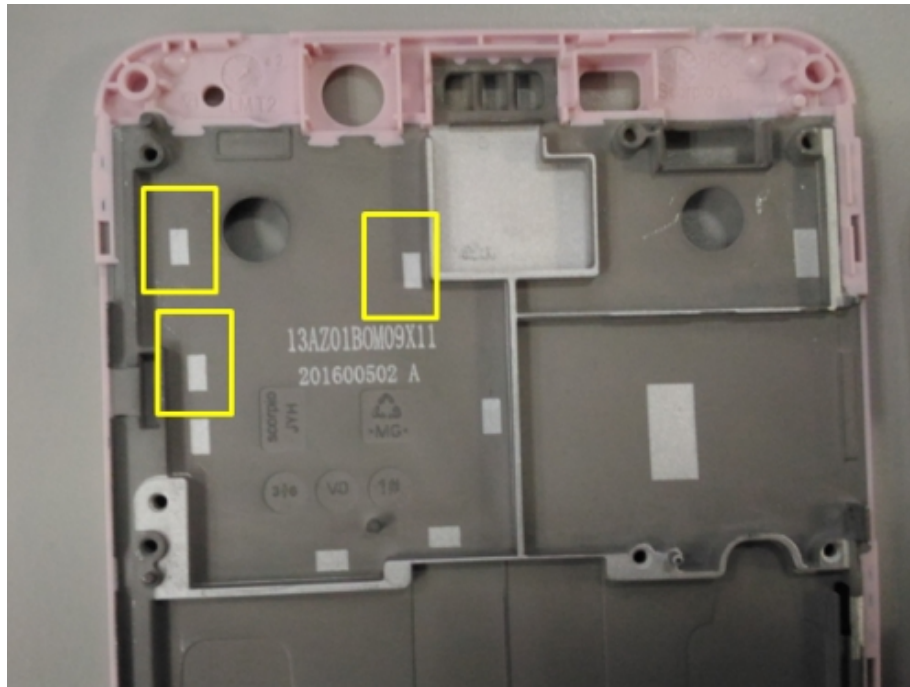
- GPS BT/Wifi test point of main board
- WIFI & GPS DIPLEXER (U5202)
- WIFI SAW FILTER (U5203)

1109 , 8072, 1127 check points 分析要點:

1. Is there a lot of flux in the test point? 檢查測試點是否有過多助焊劑
2. WIFI/BT Tx & Rx Fail
Check whether GPS function is OK or not.
 - a. GPS PASS →check WIFI /BT path
 - b. GPS FAIL →check U5202
3. Weld coaxial RF cable on the test point and connect with instrument to confirm the conducted performance

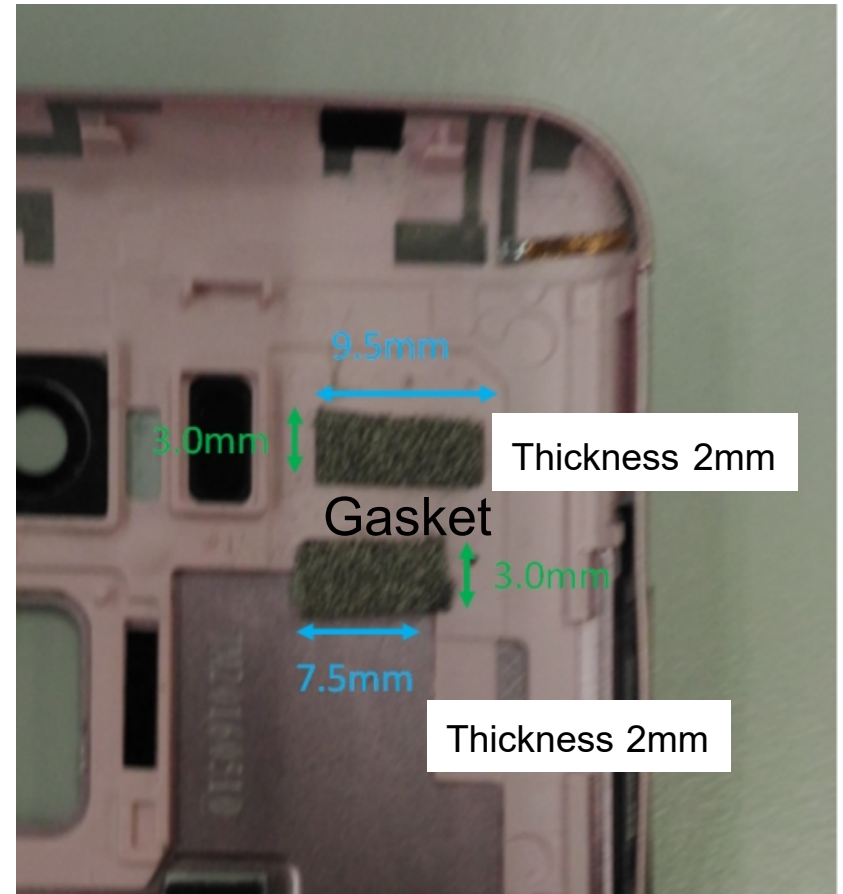
GPS Function Test Procedure

Front frame and cover are attached the gaskets shown below



Thickness 1.5mm

Gasket



GSM,WCDMA and LTE BL Tx/Rx fail analysis

2G/3G/4G Test	
2G/3G/4G TX,RX異常	
利用GOLDEN SAMPLE確認K站與測站環境正常後，重K重測。以下條列常見 error code	
1025	GSM Tx power fail
1032	GSM RSSI fail
1042	WCDMA Tx maxi. fail
1044	WCDMA SEM Mask fail
1046	WCDMA ACLR Fail，確認其對應的WCDMA maxi. Power是否過高，若過高則重K重測。
1063	LTE ACLR Fail。確認其對應的LTE maxi. Power是否過高，若過高則重K重測。
1069	LTE Main RSRP fail
1030	GSM PEAK FREQ OUT_OF_RANGE，為nonsignaling已知issue，若signaling建call測試pass spec. ，則協調RD放寬spec.。

GPS BL fail analysis

GPS Test	
GPS 異常	
利用GOLDEN SAMPLE確認K站與測站環境正常後，重K重測。以下條列常見 error code	
8205	GPS測試失敗
8206	GPS CN fail spec
1. 輸入GPS Test的ADB command或用GPS Test App確認測試環境正常 2. 確認接線上衰減器的值；GPS信號源連接的電腦重啟軟體並調到合理的輸入Power 3. 檢查原件是否有擊反，偏移，缺件等SMT問題 4. 重新放置U5302(SAW Filter)&U5303(LNA)	

Use QSPR XTT file to debug errors

1. 確認環境設定，一般設定路徑如下：

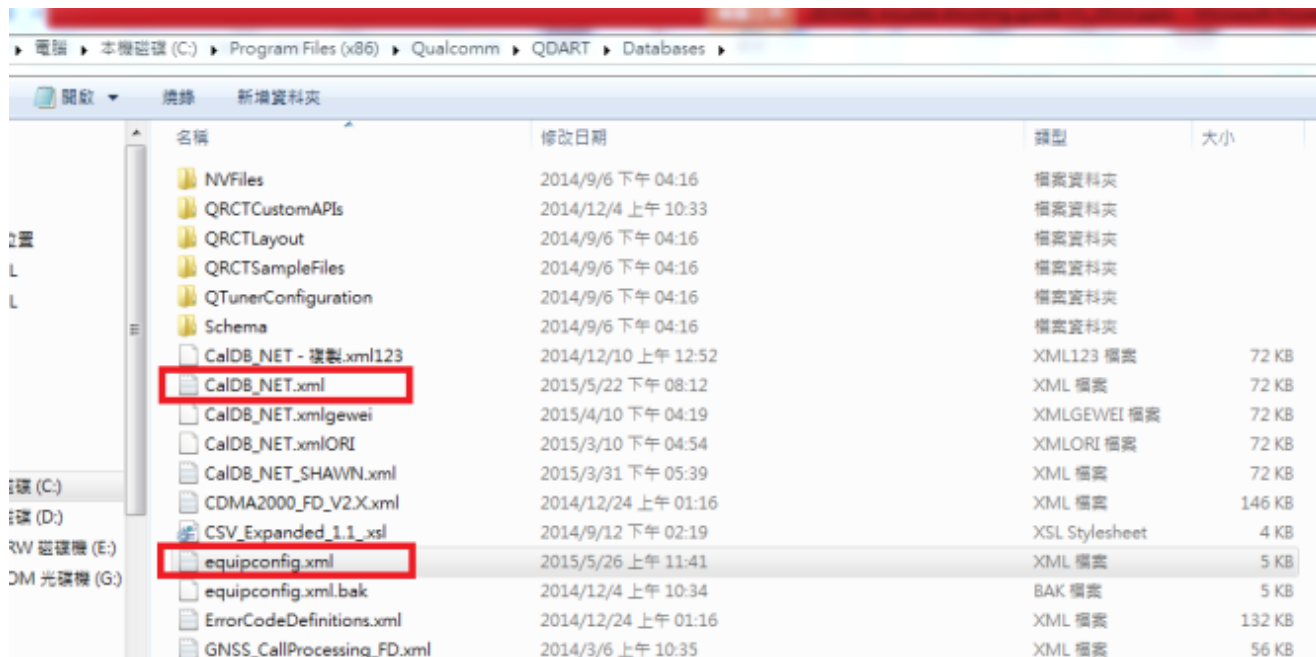
C:\Program Files (x86)\Qualcomm\QDART\Databases (64位元作業系統)

C:\Program Files\Qualcomm\QDART\Databases (32位元作業系統)

在此資料夾中有：

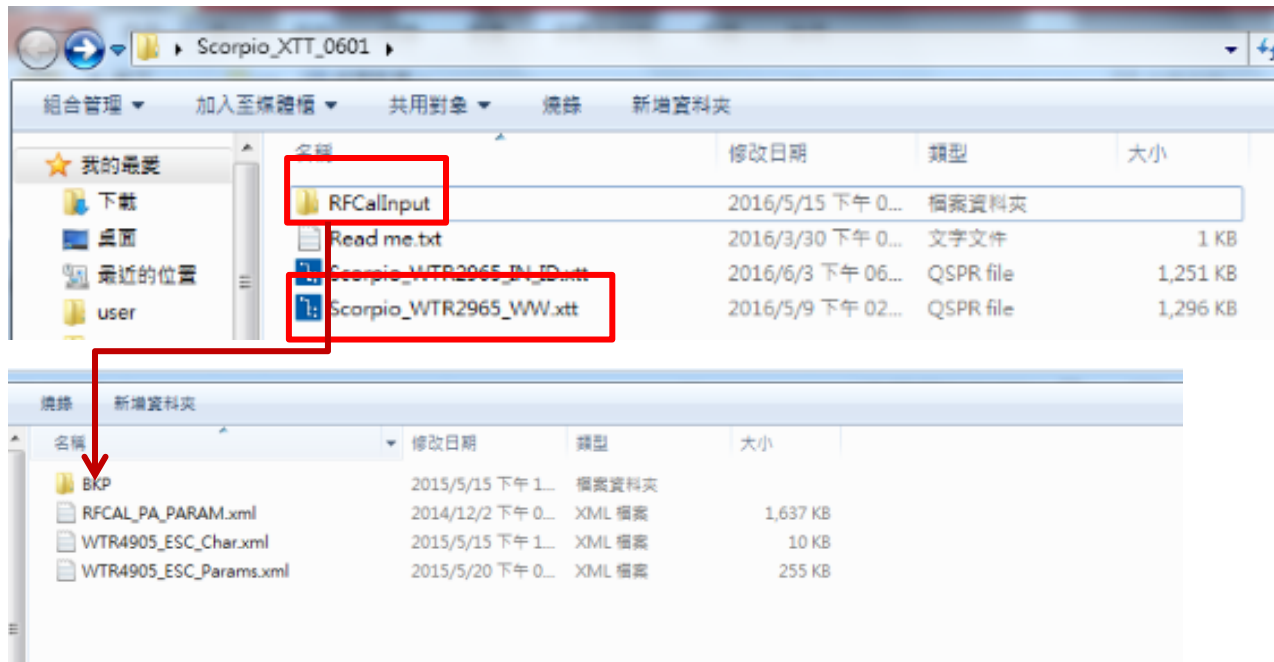
a. 進行校正(Calibration)的loss檔，RF短線的loss設0.4~0.8dB，製具壓測的loss設4~6dB。

b. equip config檔



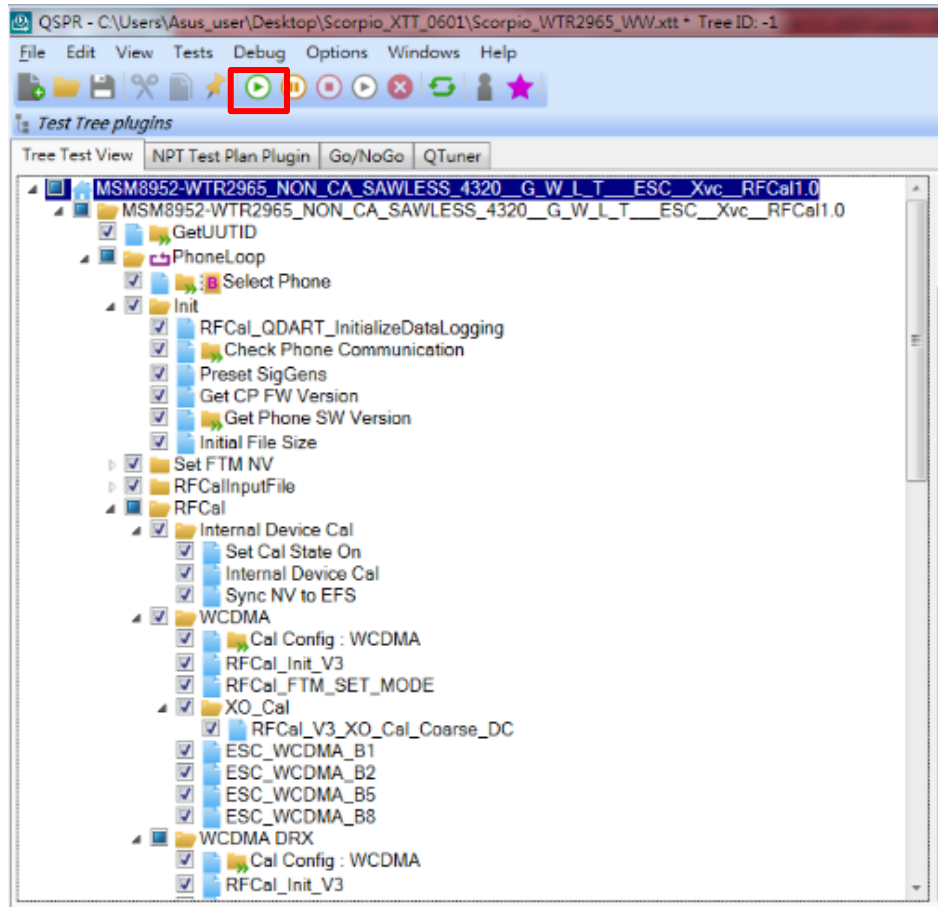
Use QSPR XTT file to debug errors

2. 確認RFCal Input File，此檔案由RD端提供，目的在定義K站的TRx規格(spec.)，如：RF Tx的PA偏壓點(VCC與ICQ參數)，PA sweep的最大功率，及Rx的LNAoffset spec.的上下限，用以驗證TRx的通透性(conductivity)、SMT有無問題。
3. 上製具壓測，廠測版image是利用USB線驅動開機，等待直到工具列右下角出現  圖示，則可以打開XTT進行校正。



Use QSPR XTT file to debug errors

4. 不同sku的板子依照其國別(WW;IN)，開啟對應的xtt檔進行RF校正，如左圖勾選，並下拉選單勾選需要校正的頻段，最後按下▶ 執行程式。

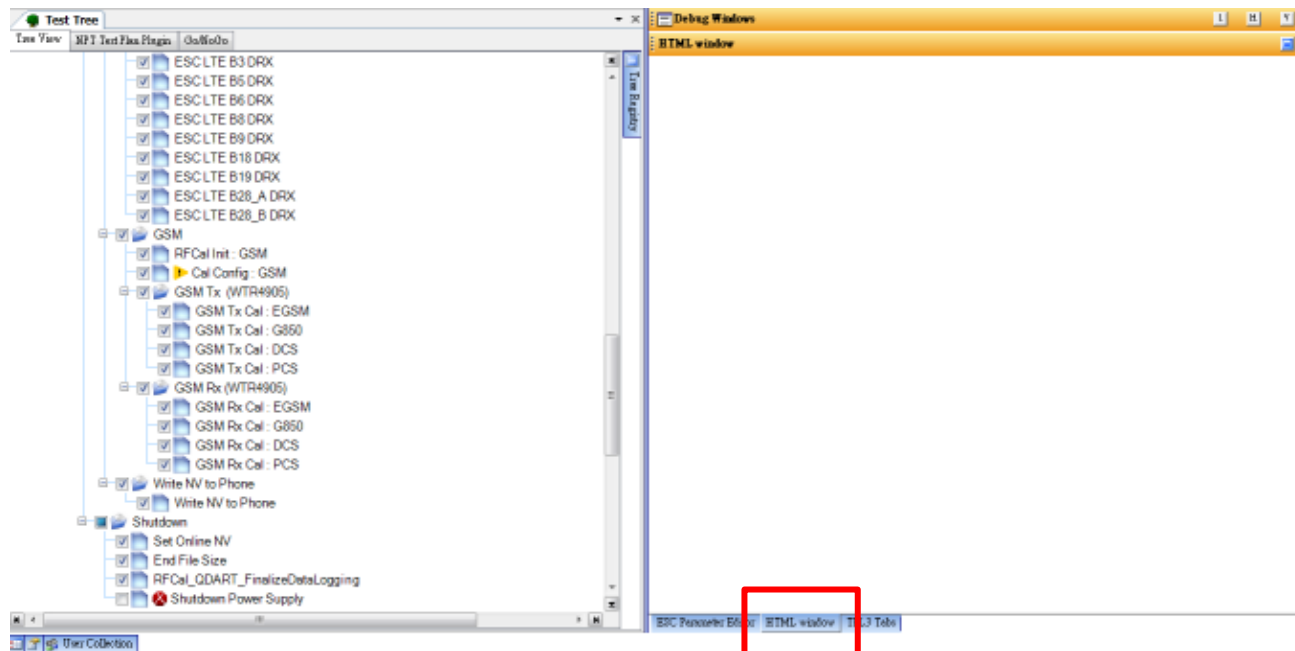


RfCalInput	2016/5/15 下午 0...	檔案資料夾	
Read me.txt	2016/3/30 下午 0...	文字文件	1 KB
Scorpio_WTR2965_IN_ID.xtt	2016/5/3 下午 06	QSPR file	1,251 KB
Scorpio_WTR2965_WW.xtt	2016/5/9 下午 02...	QSPR file	1,296 KB

Use QSPR XTT file to debug errors

5. 執行完程式後，可從K log來進行error code的硬體除錯，可以從QSPR右側分割視窗的html windows確認，或從電腦存放的K log檔，存放路徑如下：

C:\Qualcomm\QDART\Logs\Scorpio_WTR2965_WW\HTML



Use QSPR XTT file to debug errors

6. 確認K站log，如下圖所示，須注意以下幾點：

1. Tx Sweep power—確認此數值是否適當，若過大或過小則檢查該頻段所補的loss是否合理，並參考trouble shooting guide追蹤各個band的RF路徑是否有SMT問題。

2. Rx LNAOffset是否在spec.範圍內，需要TX K pass才能K Rx。

ESC_WCDMA_B1

Status: PASS

Run Info

Temp (C)	RFMode	Technology	RF Mode	Band	Device	Instance	Calibration Mode	Time (s)
22	9	WCDMA	9	1	0	0	Composite	4.49

Segment Lengths

Tx Lin Tx Seg Len (ms)	TxRx Seg Len (ms)	RF Config Seg Len (ms)	Tuning Seg Len (ms)	Time (s)
8	8	120	8	

MEASUREMENTS: Tx Linearizer Sweep

Channel	PA State	TxAGC	Vcc	Temp	Power	DET	LPM HDET	DeltaPwr	DeltaPwr Min	Time (s)
9750	1	80	3500	127	27.6	65335	-	0	0	
9750	1	79	3500	127	27.5	65335	-	0.1	-1	
9750	1	78	3500	127	27.4	63710	-	0.1	-1	
9750	1	77	3500	127	27.2	61159	-	0.2	-1	
9750	1	76	3500	127	27	58223	-	0.2	-1	
9750	1	75	3500	127	26.5	52133	-	0.5	-1	
9750	1	74	3500	127	26	47021	-	0.5	-1	
9750	1	73	3500	127	25.3	39990	-	0.7	-1	
9750	1	72	3500	127	24.5	33974	-	0.8	-1	
9750	1	71	3500	127	23.6	28029	-	0.9	-1	
9750	1	70	3500	127	22.5	21739	-	1.1	-1	
9750	1	69	3500	127	21.3	16764	-	1.2	-1	
9750	1	68	3500	127	20.3	13151	-	1	-1	
9750	1	67	3500	127	19.2	10369	-	1.1	-1	
9750	1	66	3500	127	18.2	8206	-	1	-1	

MEASUREMENTS: LNA Offset Freq Comp Device: 0

RxPath	RFMode	Channel	RxLevel (dBm)	LNARange	RxPM	RxFCompLNAOffset	RxFCompLNAOffset Min	RxFCompLNAOffset Max
0	9	9613	-70	0	0	-95	-150	-40
0	9	9613	-60	1	0	86	40	150
0	9	9613	-50	2	0	277	220	330
0	9	9613	-42	3	0	365	300	410
0	9	9633	-70	0	0	-96	-150	-40
0	9	9633	-60	1	0	86	40	150
0	9	9633	-50	2	0	274	220	330
0	9	9633	-42	3	0	366	300	410
0	9	9653	-70	0	0	-96	-150	-40
0	9	9653	-60	1	0	87	40	150
0	9	9653	-50	2	0	276	220	330
0	9	9653	-42	3	0	368	300	410
0	9	9673	-70	0	0	-96	-150	-40
0	9	9673	-60	1	0	87	40	150
0	9	9673	-50	2	0	277	220	330
0	9	9673	-42	3	0	370	300	410
0	9	9693	-70	0	0	-95	-150	-40
0	9	9693	-60	1	0	86	40	150
0	9	9693	-50	2	0	277	220	330